COS 126

Introduction To Computer Science

Ruth Fong



PRINCETON
UNIVERSITY

Introduction to Machine Learning

Exciting developments in machine learning today

What are some exciting developments you've heard of in machine learning?

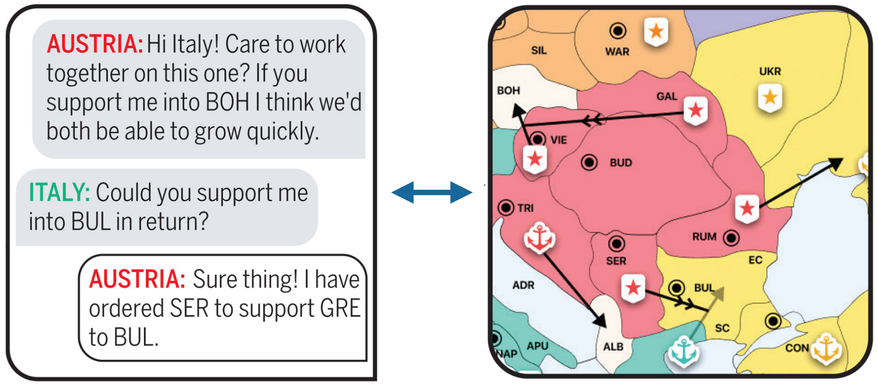
MI How are you today, ChatGPT?

As a language model trained by OpenAI, I don't have the ability to experience emotions like a human would. I am designed to assist with providing information and answering questions to the best of my ability. Is there something specific you would like to know?

MI What is the Capital of the Klingon homeworld, Qo'noS?

In the fictional Star Trek universe, the Klingon homeworld is called Qo'noS and its capital is called First City. It is the political and cultural center of the Klingon Empire and the location of the Klingon High Council. It is a city of great size and importance in Klingon society.

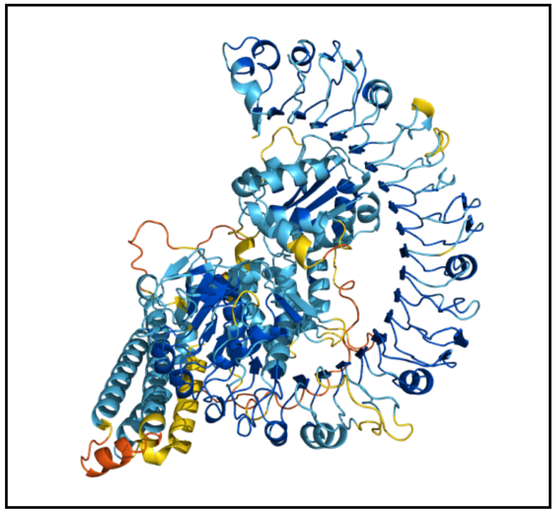
ChatGPT



Cicero



DALLE-2



AlphaFold

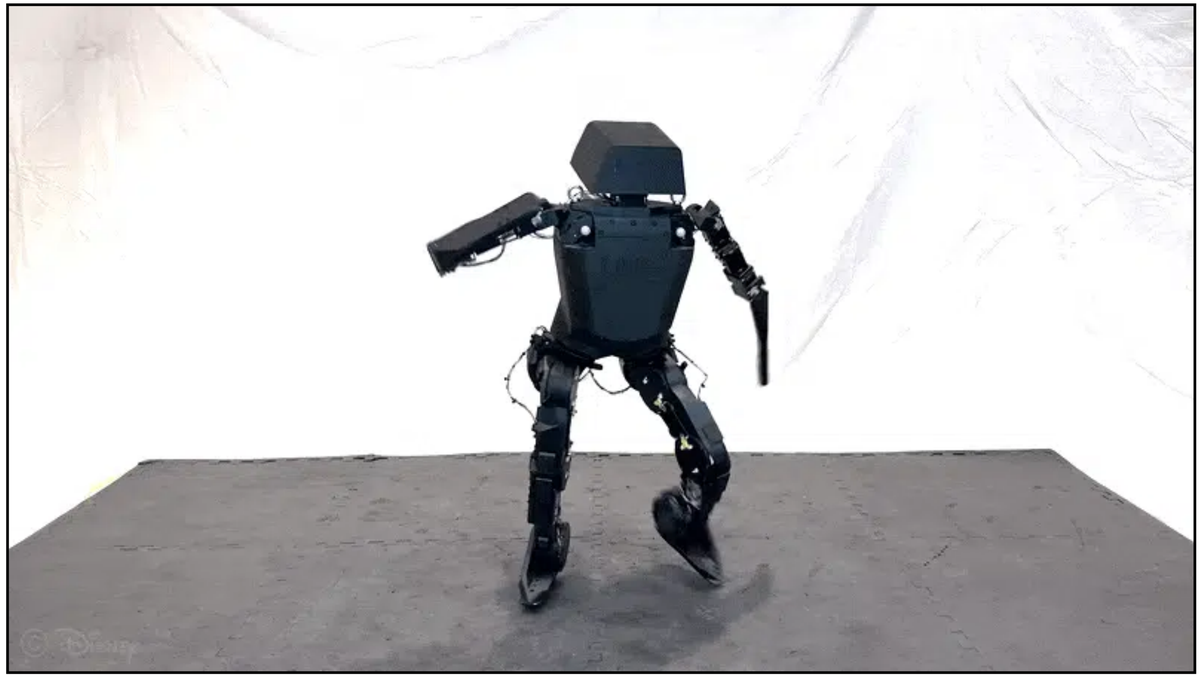
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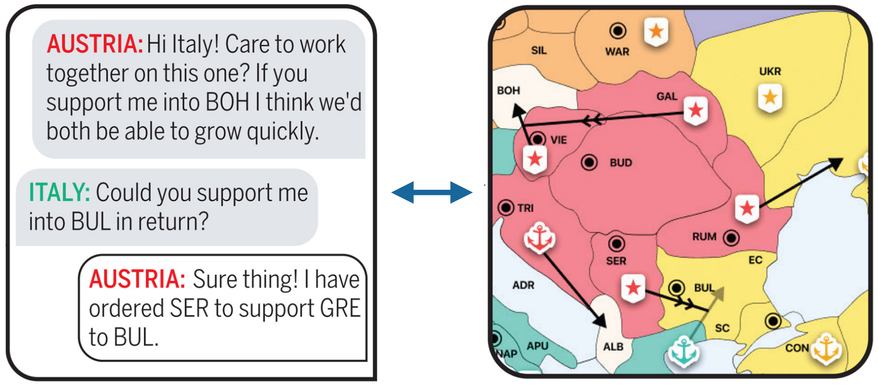
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ChatGPT



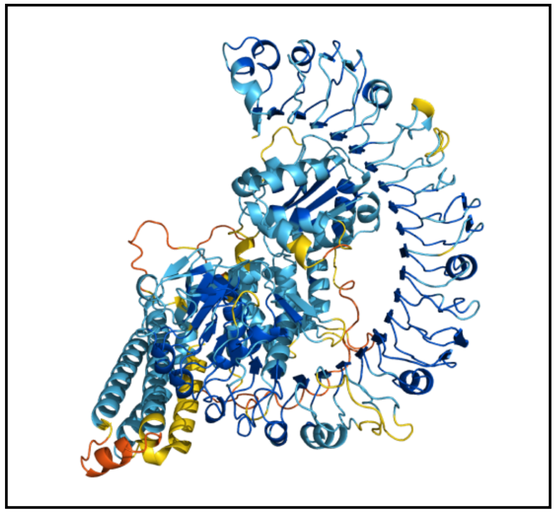
Disney Research



Cicero



DALLE-2



AlphaFold

Roadmap

- What is machine learning? How does it relate to other fields?
- Machine learning paradigms
  - Supervised learning
  - Unsupervised learning
  - Reinforcement learning
- Perceptron
  - Example: Golden retriever vs. Doberman
  - Perceptron model
  - Training via update rule
  - Testing
- Multi-layer Perceptron (MLP)
- Multi-class Perceptron

Introduction to machine learning

ML data ⇔ related fields

Player	Minutes	Points	Rebounds	Assists
A	41	20	6	5
B	30	29	7	6
C	22	7	7	2
D	26	3	3	9

Tabular

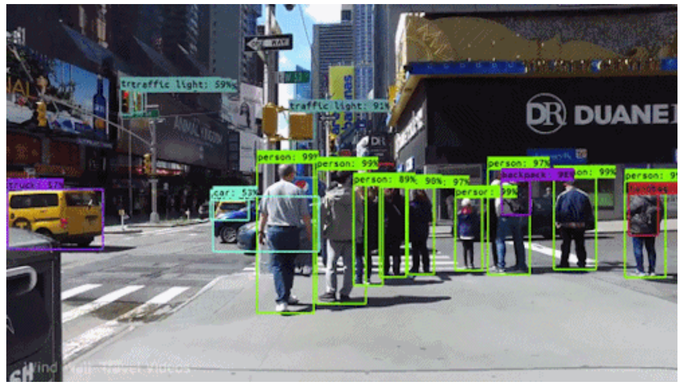
Visual

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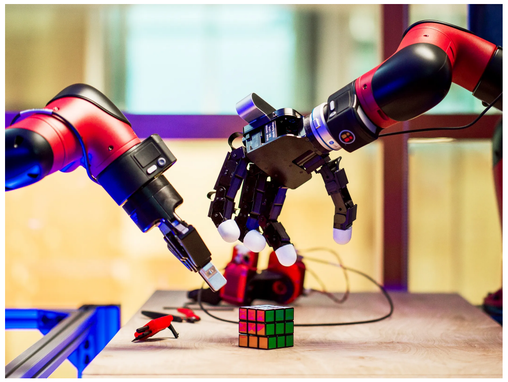
Tabular

Machine learning



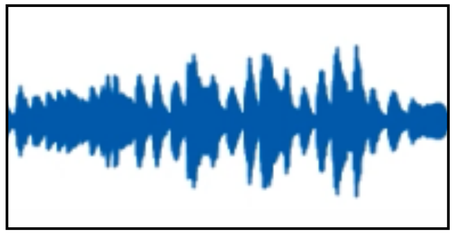
Visual

Computer vision



Real manipulation

Robotics



Audio

Audio processing

ES

Briefly explain chatGPT using a metaphor that would resonate with an audience of stock-market investors and avoiding AI jargon.

One way to explain chatGPT to stock-market investors is to compare it to a financial advisor who uses past performance data and market trends to make predictions about future investments. Just like a skilled financial advisor, chatGPT uses large amounts of training data to generate intelligent and informed responses to questions, providing valuable insights and suggestions to users.

Text

Natural language processing (NLP)

What is machine learning?

Machine learning: Algorithms that improve automatically using data or repeated experience.

This was **not** true of the programs that you written so far in COS1 26
(e.g. your programs = explicit instructions for the computer).

Note: There are non-machine learning algorithms in computer vision, NLP, robotics, etc.

Machine learning paradigms

1. Supervised learning: learn to **predict** an output from an input.
2. Unsupervised learning: learn **patterns** from data.
3. Reinforcement learning: learn by **interacting** (i.e. **trial-and-error**) with an environment.

Supervised learning

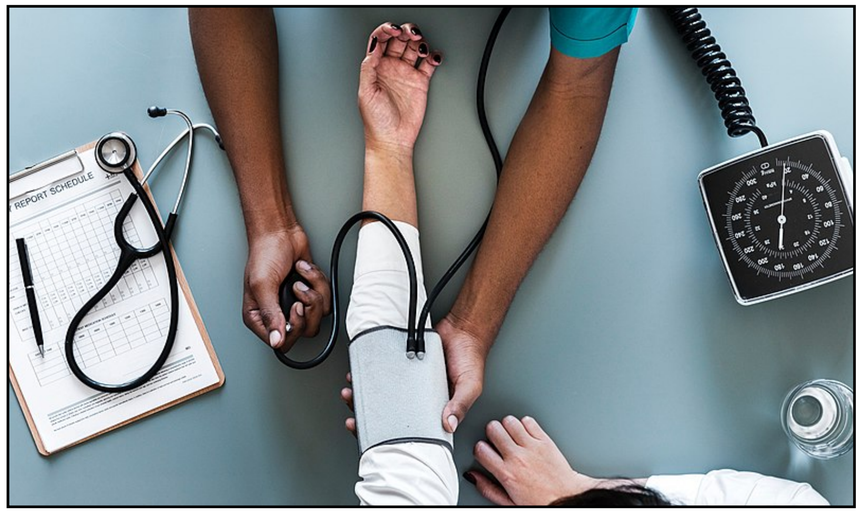
Given (input, output) pairs, **predict** an output from its input.

$\text{input}_1 \rightarrow \text{output}_1$
$\text{input}_2 \rightarrow \text{output}_2$
$\text{input}_3 \rightarrow \text{output}_3$

Supervised learning

Medicine

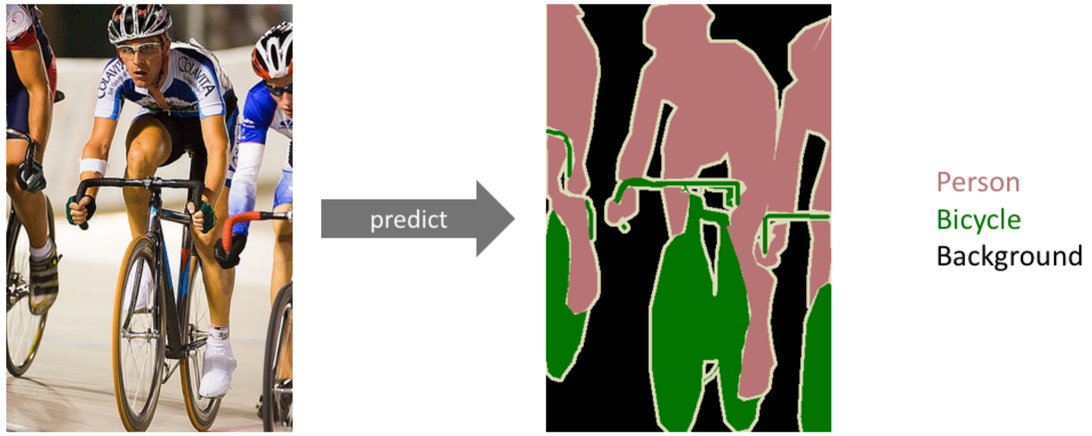
- Input: medical information (i.e. age, body mass index [BMI], blood sugar levels, etc.)
- Output: blood pressure (in mm Hg)



Supervised learning

Computer vision: segmentation

- Input: an image
- Output: segmentation map of objects

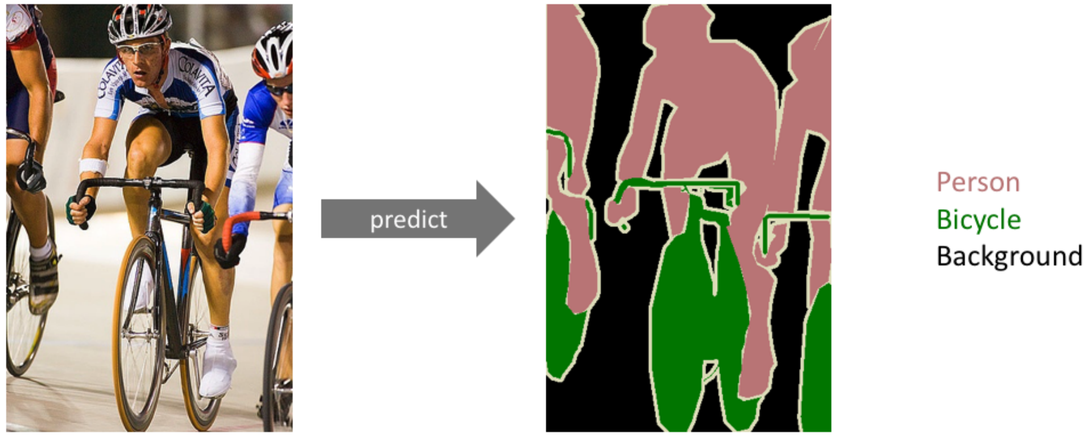


Pascal VOC dataset
http://host.robots.ox.ac.uk/pascal/VOC/voc2012/

Supervised learning

Computer vision: segmentation

- Input: an image
- Output: segmentation map of objects
- **Application: self-driving cars**



Pascal VOC dataset
http://host.robots.ox.ac.uk/pascal/VOC/voc2012/



https://github.com/meetps/pytorch-semseg

Supervised learning

NLP: Machine translation

- Input: a sentence in English
- Output: its translation in Spanish
- **Application: Google Translate**



Gender Bias in Machine Translation:
https://towardsdatascience.com/gender-bias-in-machine-translation-819ddce2c452

"That river is dangerous to swim in." → "Es peligroso nadar en ese río."

"The baby is playing with some toys." → "El bebé juega con algunos juguetes."

Anki Tab-Delimited Bilingual Sentence Pairs
https://www.manythings.org/anki/

Supervised learning

Other applications

- Spam filtering: email text → {spam, not spam}
- Speech recognition: audio clip → text transcription
- Medical imaging: CT scan of COVID patients → severity of COVID symptoms
- …

If you can create a dataset of (input, output) pairs, you can use supervised learning.

Unsupervised learning

Given unlabelled data (i.e. no input-output pairs), learn **patterns** from data.

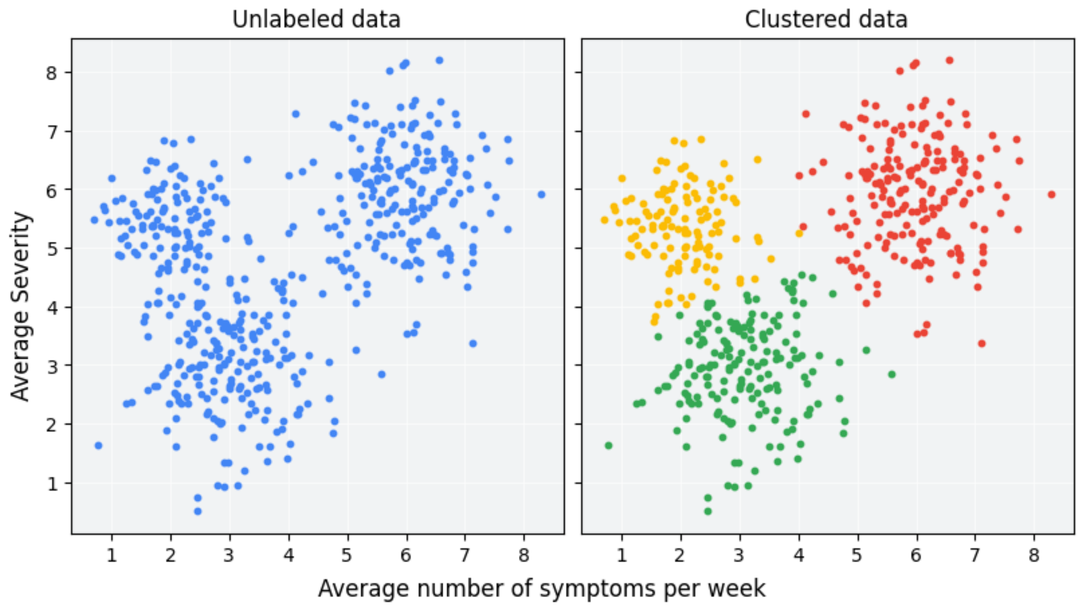
Examples:

- Clustering
- Dimensionality reduction
- Anomaly detection
- Generative modeling
- …

Unsupervised learning

Clustering

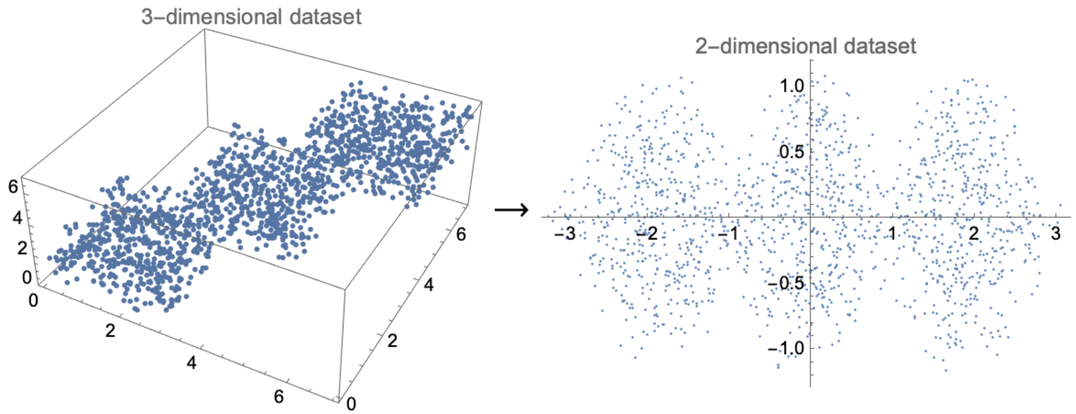
- Goal: Given a data point, categorize it into one of k clusters.
- Unlike supervised learning, there are no output labels!
  i.e. no (data point, cluster) pairs
- Application in retail marketing: What kinds of people have similar shopping habits?



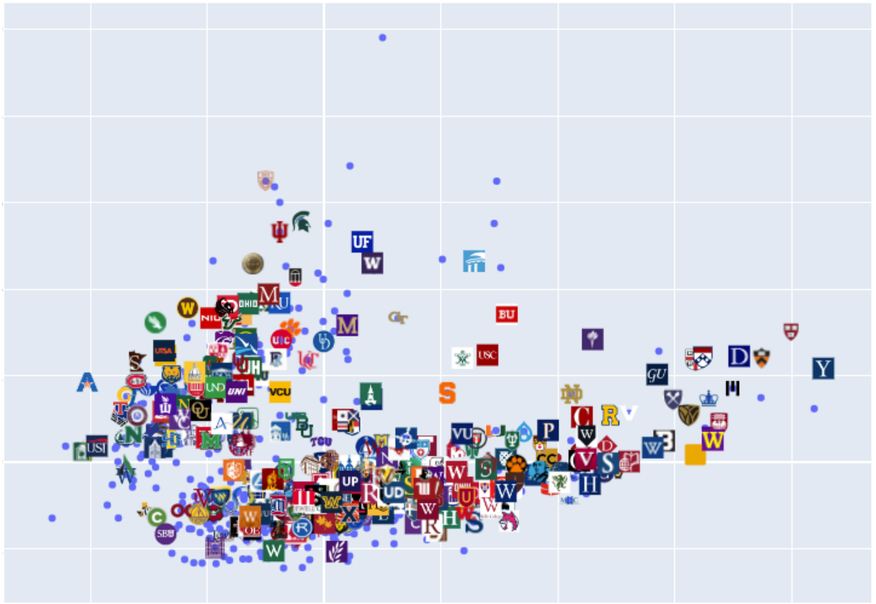
Unsupervised learning

Dimensionality reduction

- Goal: Reduce the number of variables in the data while preserving patterns in the data (e.g. distances between data points).
- Uses:
  - Compression
  - Visualization
  - …



PCA example of colleges dataset

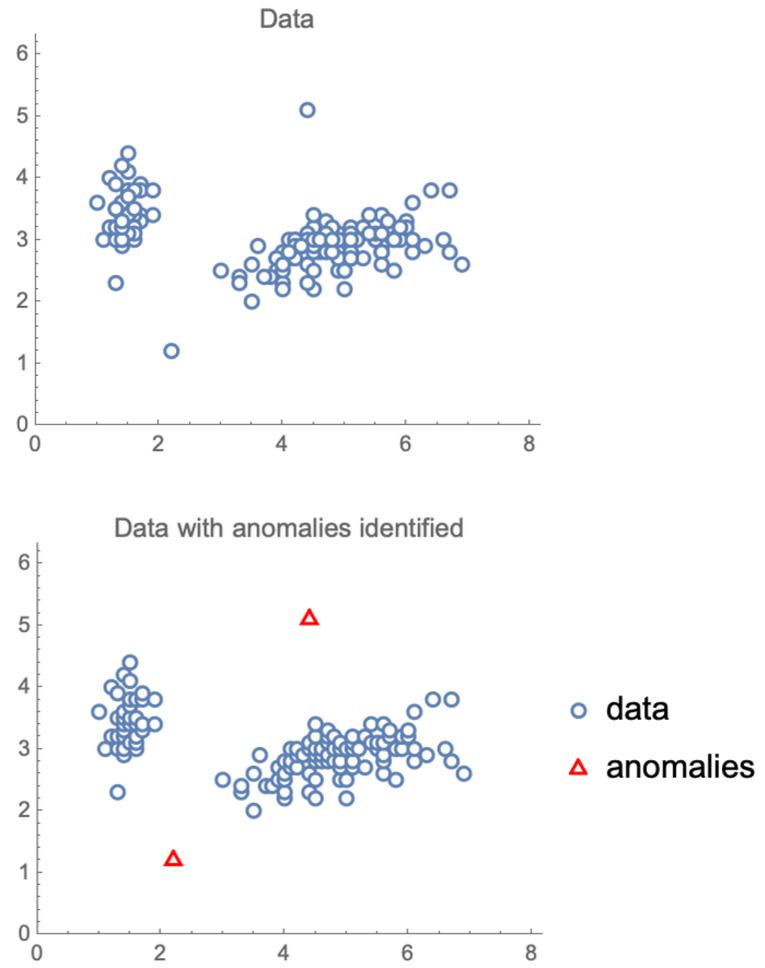


https://www.ruthfong.com/cos324/pca_colleges.html

Unsupervised learning

Anomaly detection

- Goal: Identify anomalies (i.e. outliers) in a dataset.
- Applications:
  - Credit card fraud
  - Medical diagnosis
  - …



Unsupervised learning

Generative model

- Goal: Generative synthetic examples that look realistic.
- Applications:
  - Computer vision: Given a dataset of human face images, generate realistic looking faces.
  - NLP: Given a dataset of Shakespeare's sonnets, generate a realistic sounding, Shakespearean sonnet.



Generated by StyleGAN
(Karras, et al., CVPR 2019)

"am of my faults thy sweet self dost deceive:
they look into the beauty of thy deeds
nativity, once in the chronicle of wasted time
not from the thing they see"

Generated by Markov Chains
https://rpubs.com/malcolmbarrett/shakespeare

Reinforcement learning

Learn by **interacting** (i.e. **trial-and-error**) with an *environment.*

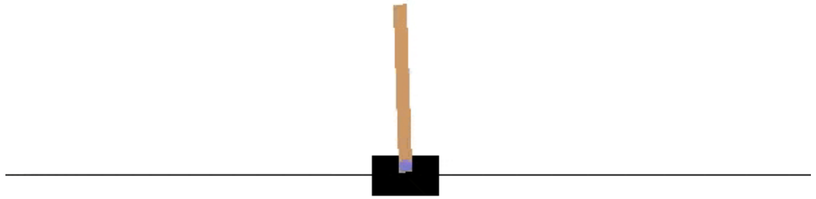
- An *agent* interacts with an *environment* by performing *actions* and then gathers *observations* (and sometimes *rewards*) that occur in response to its actions
- Interactions generate data (i.e. no fixed dataset).
- Goal: Learn *agent* to maximize *reward*.



Reinforcement learning

CartPole

- Goal: Balance a pole on a cart by moving it left and right.
- Actions: {left, right}
- Reward: +1, for each step the pole is upright.
- Learn over many attempts (i.e. episodes).

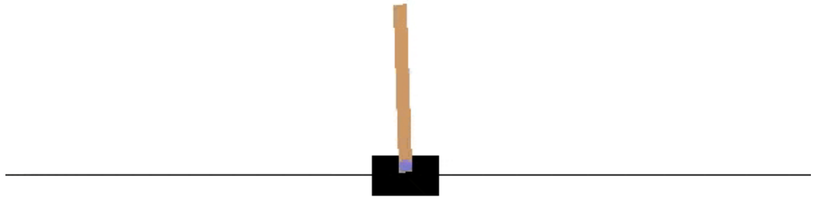


https://gymnasium.farama.org/environments/classic_control/cart_pole/

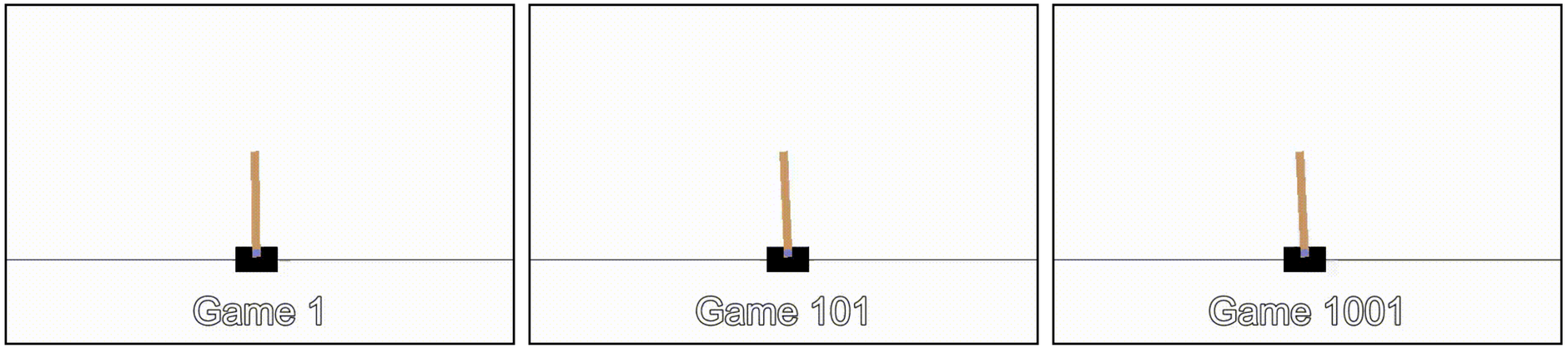
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https://blog.ml6.eu/importance-of-data-visualization-for-ml-936239c355d0

Reinforcement learning

Application: Games

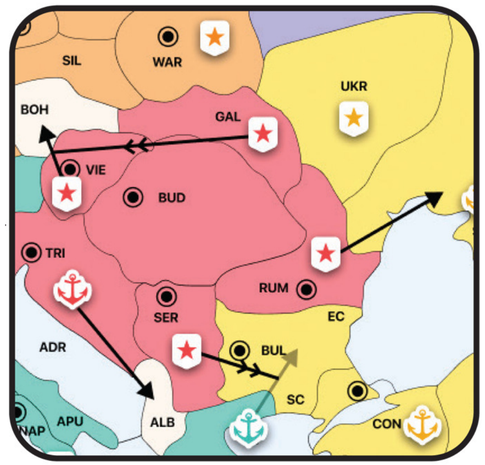
Reinforcement learning

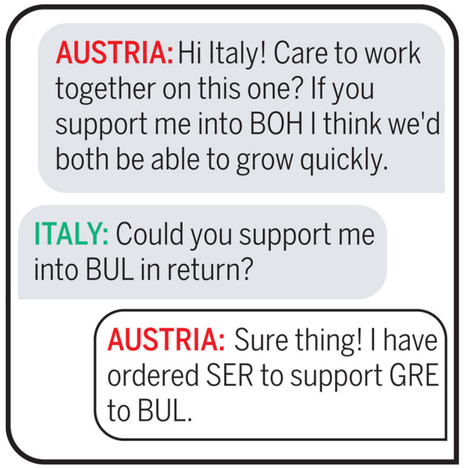
Application: Games

- Atari: DeepMind (2015)
- Go (strategy): AlphaGo from DeepMind (2016)
- Poker (multi-player): Libratus and Pluribus from CMU (2017, 2019)
- Diplomacy (cooperative): Cicero from Meta AI (2022)









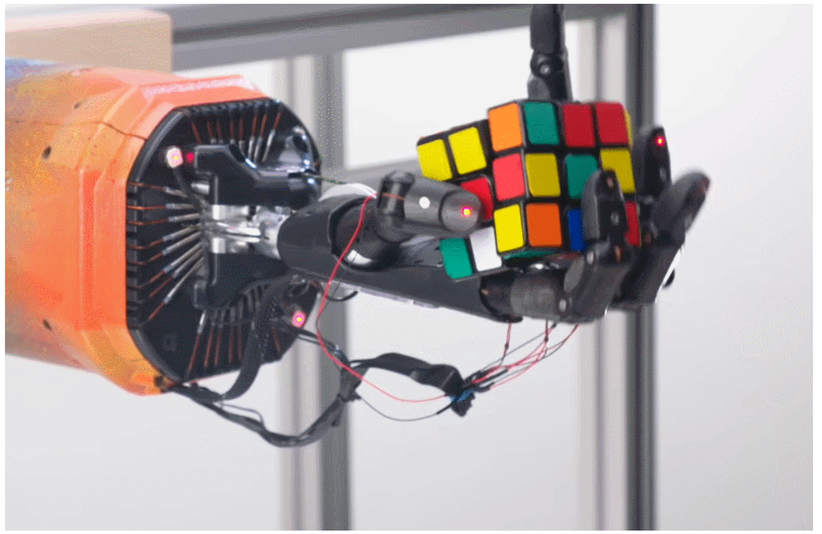
Reinforcement learning

Application: Robotics

Reinforcement learning

Application: Robotics

- Solving a Rubik's cube

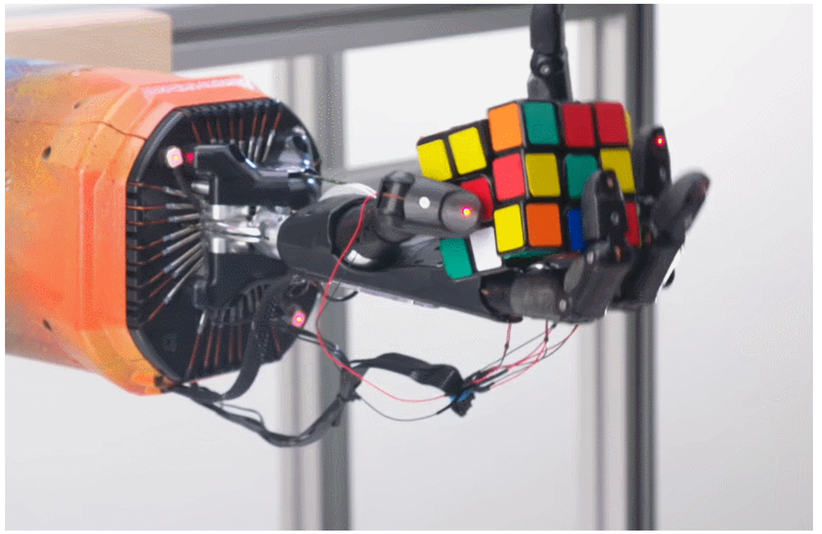


OpenAI's ADR (2019)

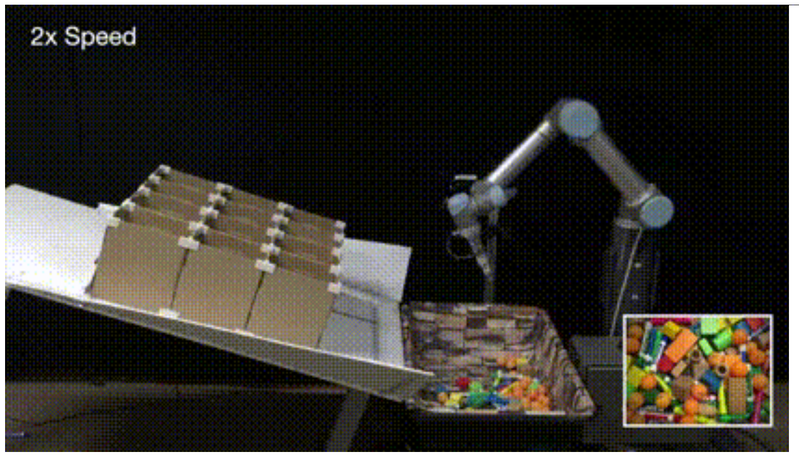
Reinforcement learning

Application: Robotics

- Solving a Rubik's cube
- Manipulating objects (e.g. picking up, throwing, etc.)



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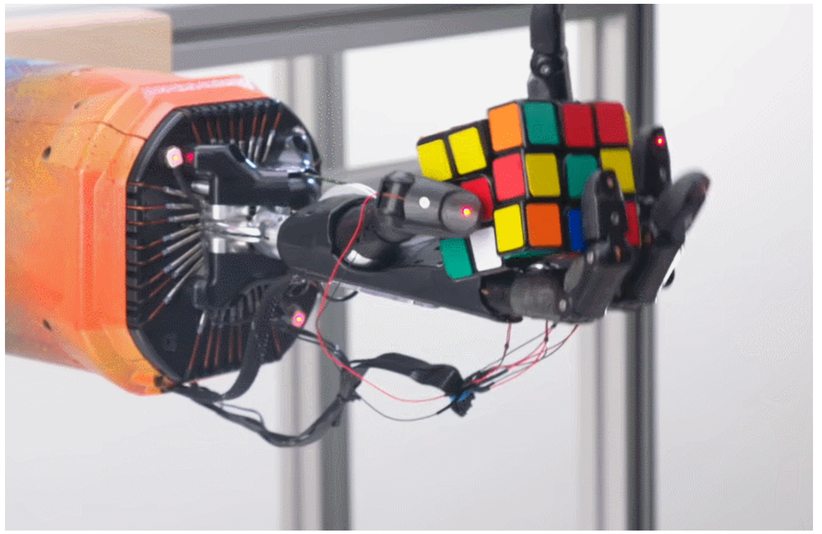


Princeton+Google's TossingBot (2019)

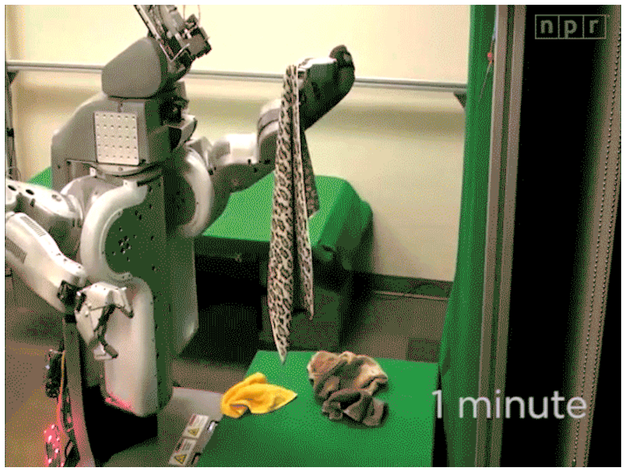
Reinforcement learning

Application: Robotics

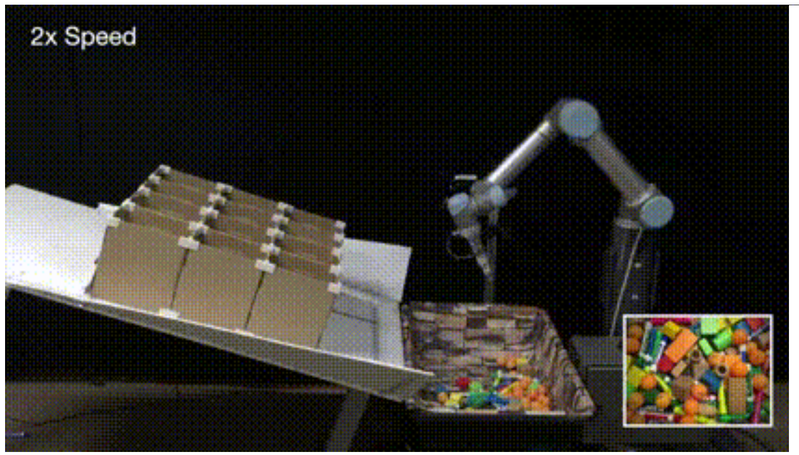
- Solving a Rubik's cube
- Manipulating objects (e.g. picking up, throwing, etc.)
- Laundry folding



OpenAI's ADR (2019)



Berkeley's BRETT (2015)



Princeton+Google's TossingBot (2019)

Poll

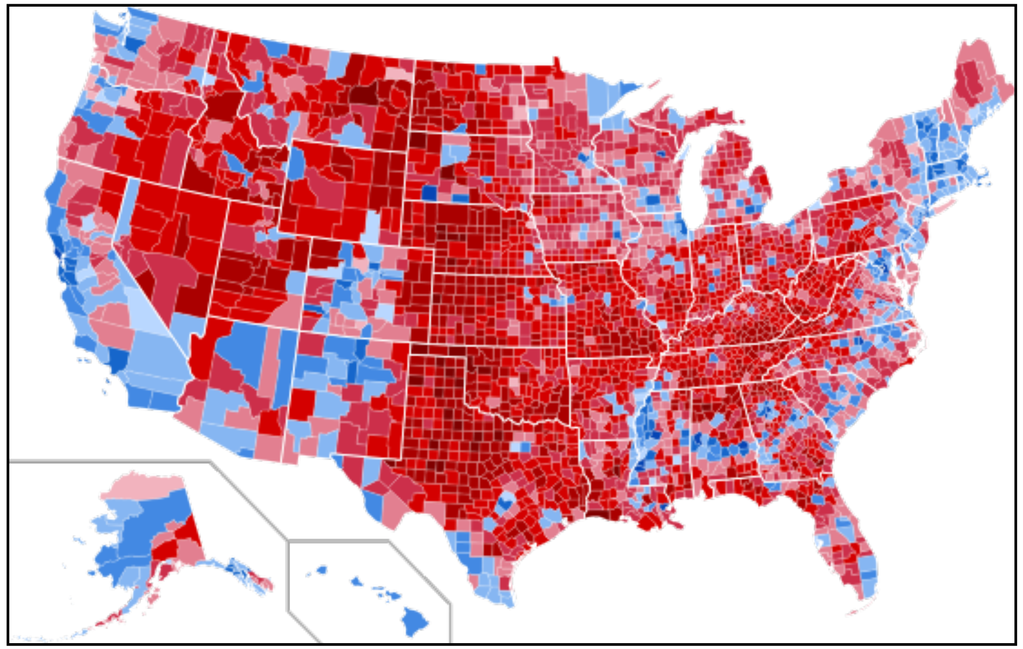
I would like to predict the outcome of the 2024 U.S. presidential election based on polling data leading up to the election and data from prior presidential elections.

What kind of machine learning problem is this?

a. Supervised learning

b. Unsupervised learning

c. Reinforcement learning



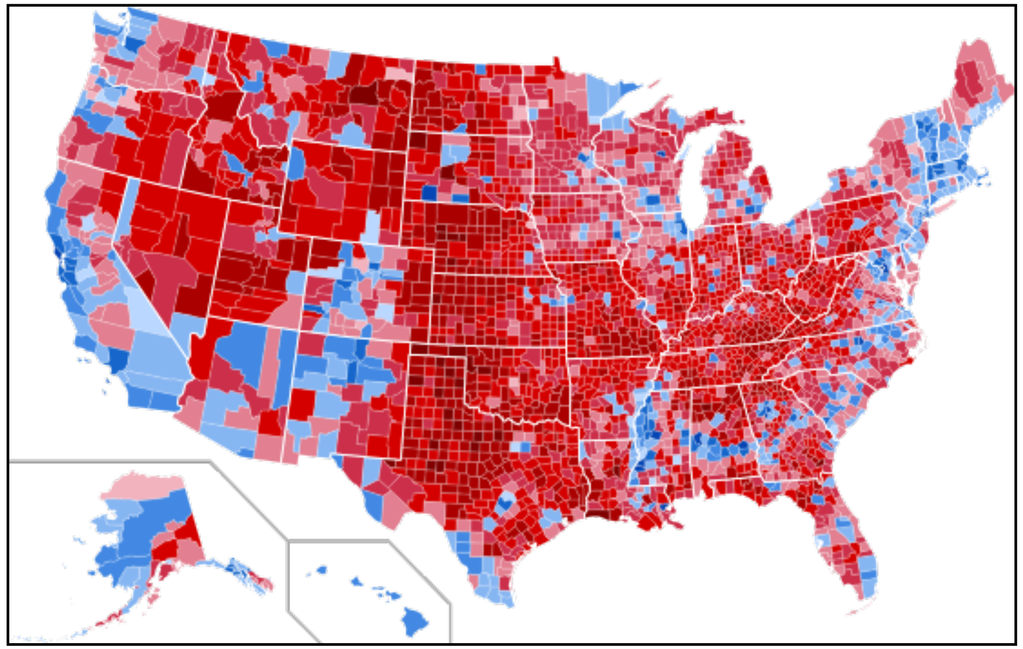
https://commons.wikimedia.org/wiki/File:2020_United_States_presidential_election_results_map_by_county.svg



Poll

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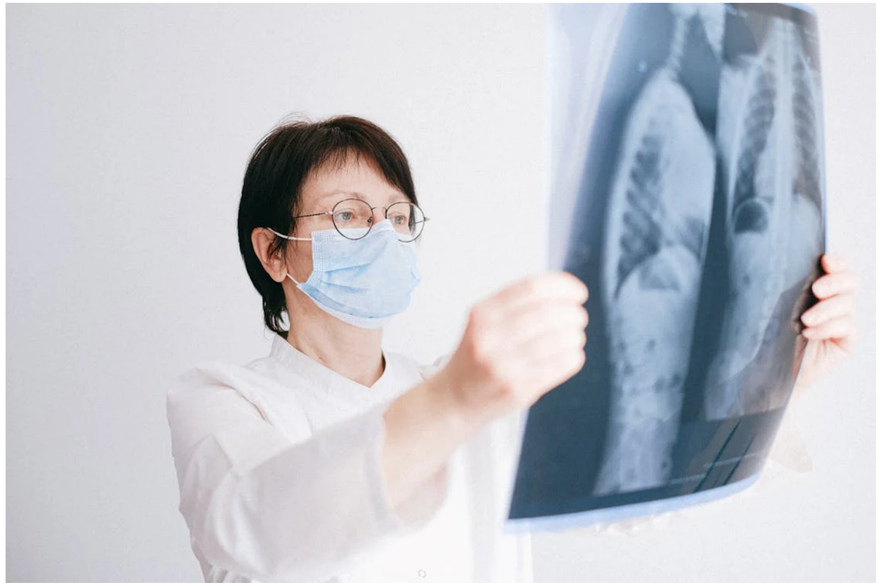
Given a dataset of lung scans of patients, I would like to predict if a new lung scan is abnormal.

What kind of machine learning problem is this?

a. Supervised learning

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c. Reinforcement learning

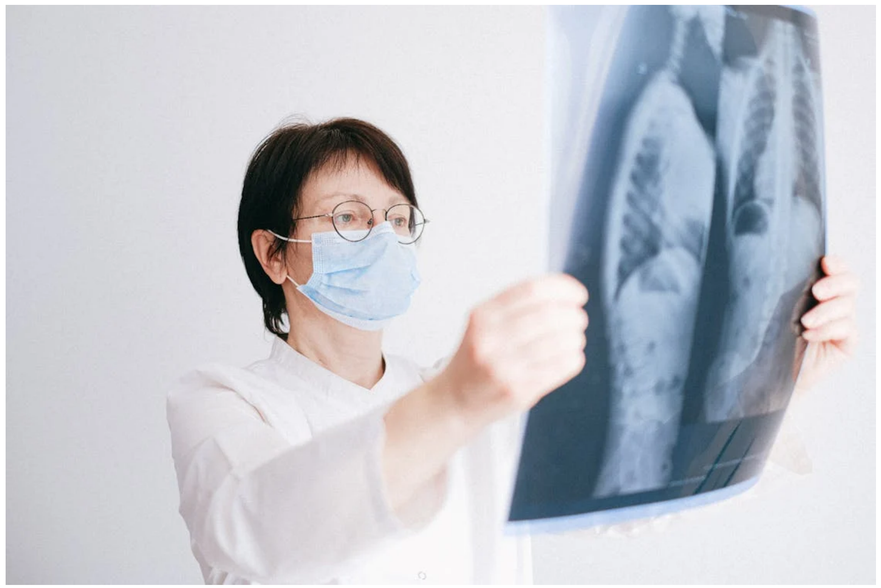


https://www.pexels.com/photo/doctor-looking-at-lung-scans-she-is-holding-4225926/

Poll

Given a dataset of lung scans of patients, I would like to predict if a new lung scan is abnormal.

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https://www.pexels.com/photo/doctor-looking-at-lung-scans-she-is-holding-4225926/

Machine learning paradigms

1. Supervised learning: learn to **predict** an output from an input.
2. Unsupervised learning: learn **patterns** from data.
3. Reinforcement learning: learn by **interacting** (i.e. **trial-and-error**) with an environment.

Other paradigms:

- *Self-supervised learning*
- *Semi-supervised learning*
- *Transfer learning*
- *Online learning*
- *…*

Ingredients of ML

1. **Data.** Can be labelled, unlabelled, or from experience.
2. **Model.** A model maps from datapoint to a desired answer or output.
3. **Model parameters**. Internal parameters of model that are learnable.
4. **Training.** Given datapoints, find good model parameters.
5. **Testing.** Evaluate the performance of learned model on new, previously unseen data (i.e. test set).

Perceptron

Supervised learning

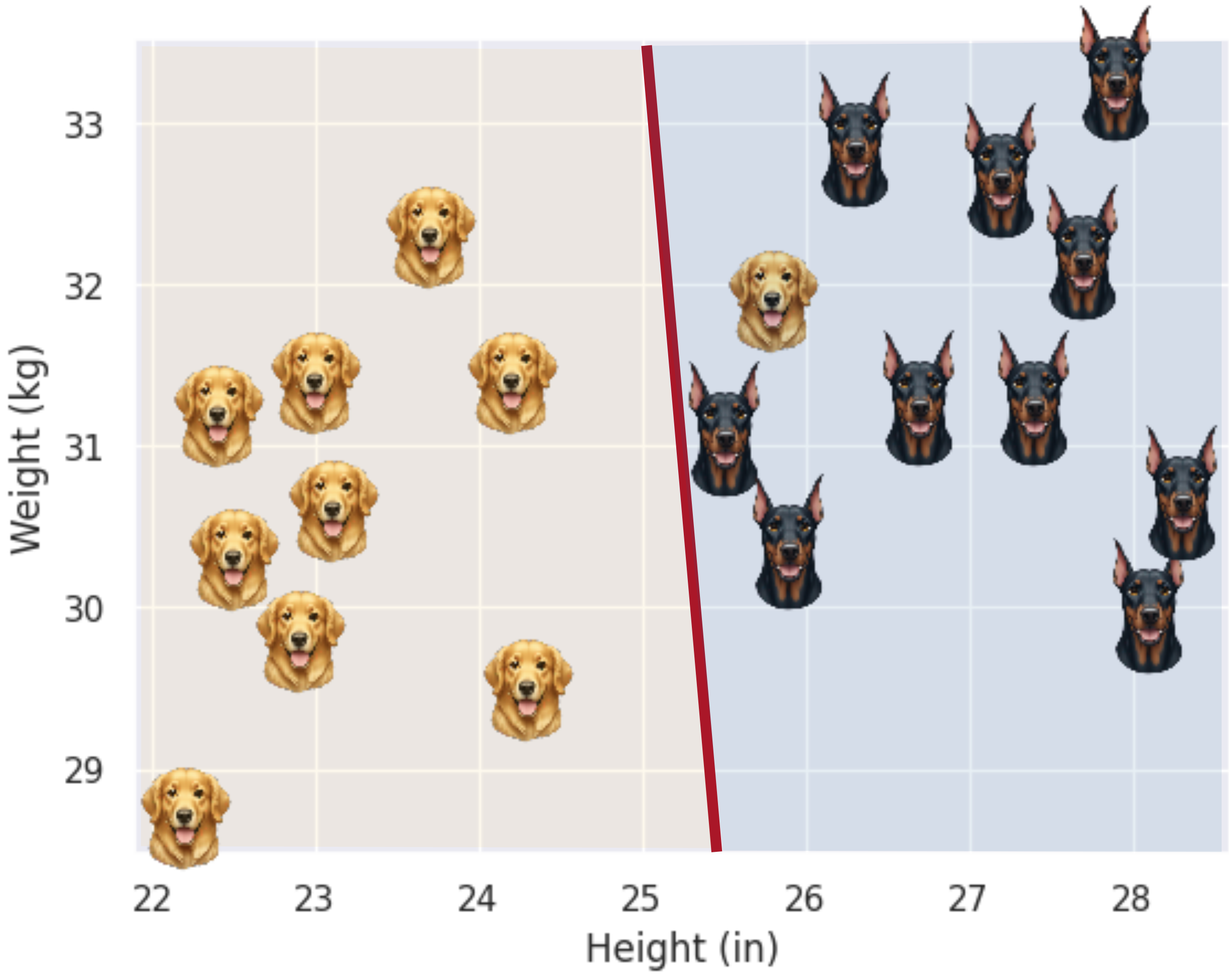
Given (input, output) pairs, **predict** an output from its input.

$$\text{input}_1 \rightarrow \text{output}_1$$
$$\text{input}_2 \rightarrow \text{output}_2$$
$$\text{input}_3 \rightarrow \text{output}_3$$

Basic problems described by output type:

- **Regression**: Predict a continuous value (e.g. number).
- **Classification**: Predict a discrete category (i.e. classes).
  - **Binary classification**: Predict one of **two** classes.
  - **Multi-class classification**: Predict one of **k** classes ($k > 2$).

Binary classification

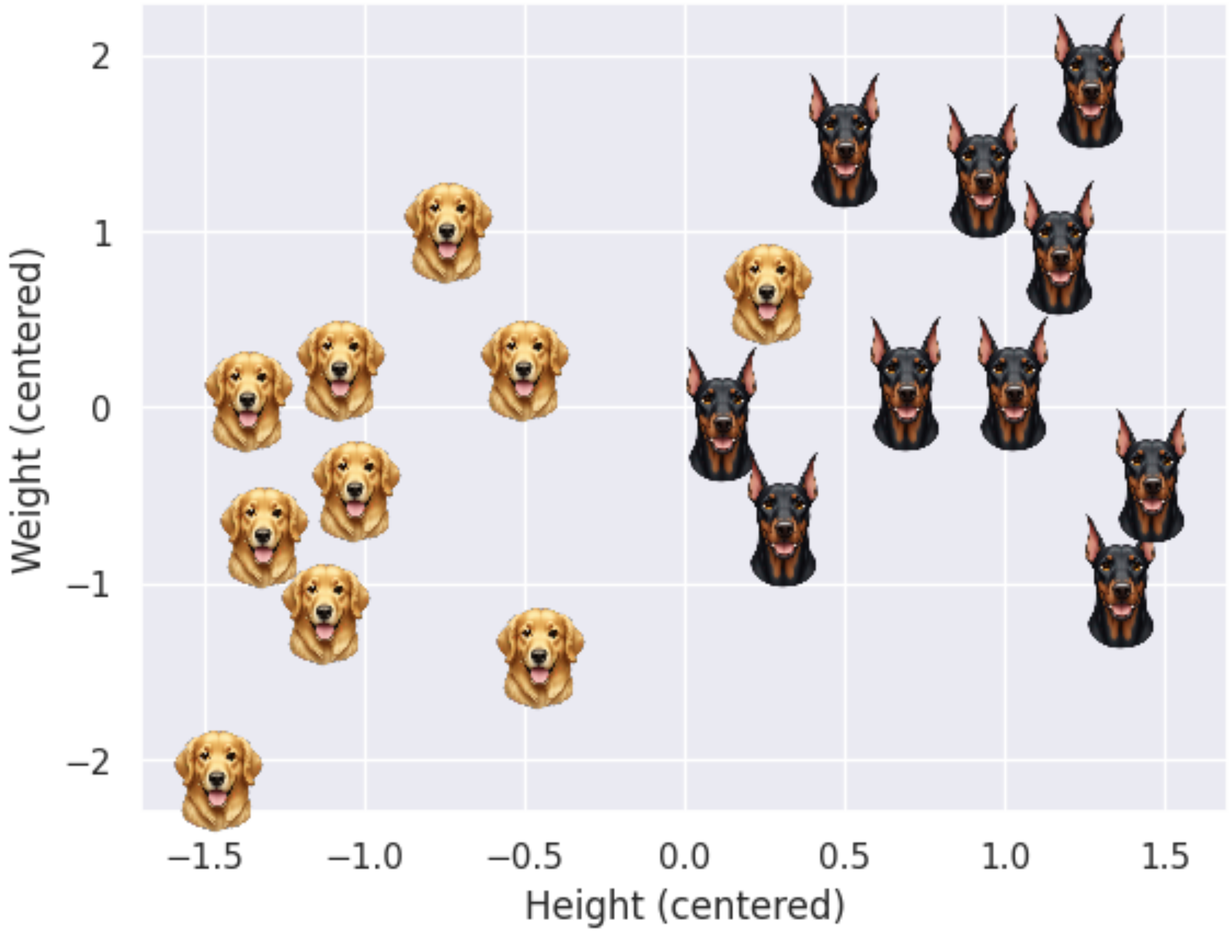


Data is nearly **linearly separable**.

Binary classification

Predict dog species

- Input (centered*)
  - Height
  - Weight
- Output
  - Golden Retriever: $+1$
  - Doberman Pinscher: -1



* Input centered to have mean = 0 and standard deviation = 1.

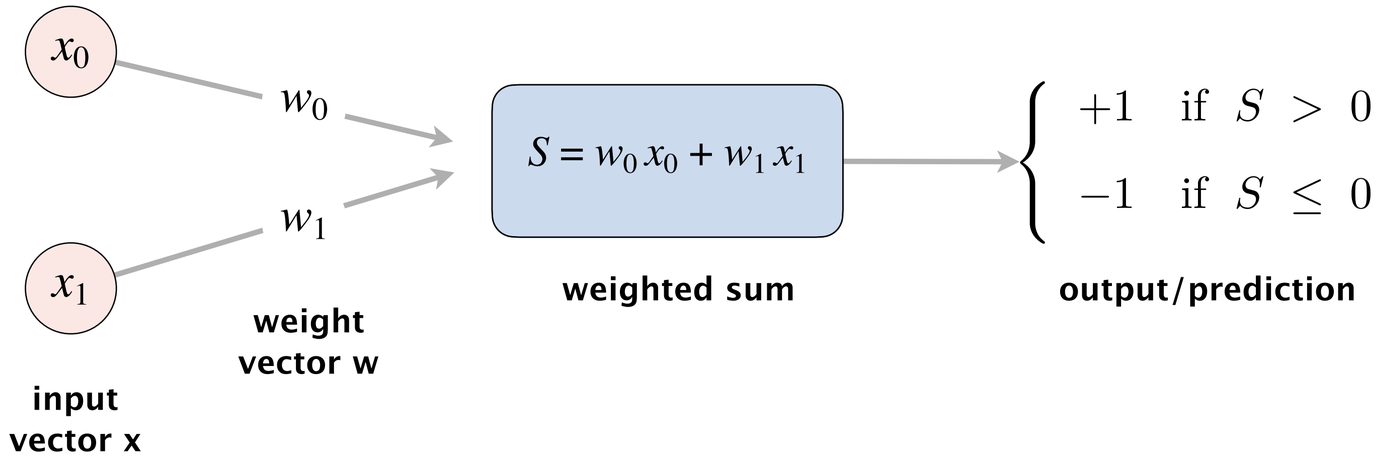
Perceptron

Algorithm that takes several inputs and produces a single, binary output.

- Inputs: x_0, x_1 (i.e. height, weight)

- Compute weighted sum
  $S = w_0 x_0 + w_1 x_1$

- Output
  $\text{sign}(S) = +1 \text{ if } S > 0$
  $= -1 \text{ if } S \leq 0$



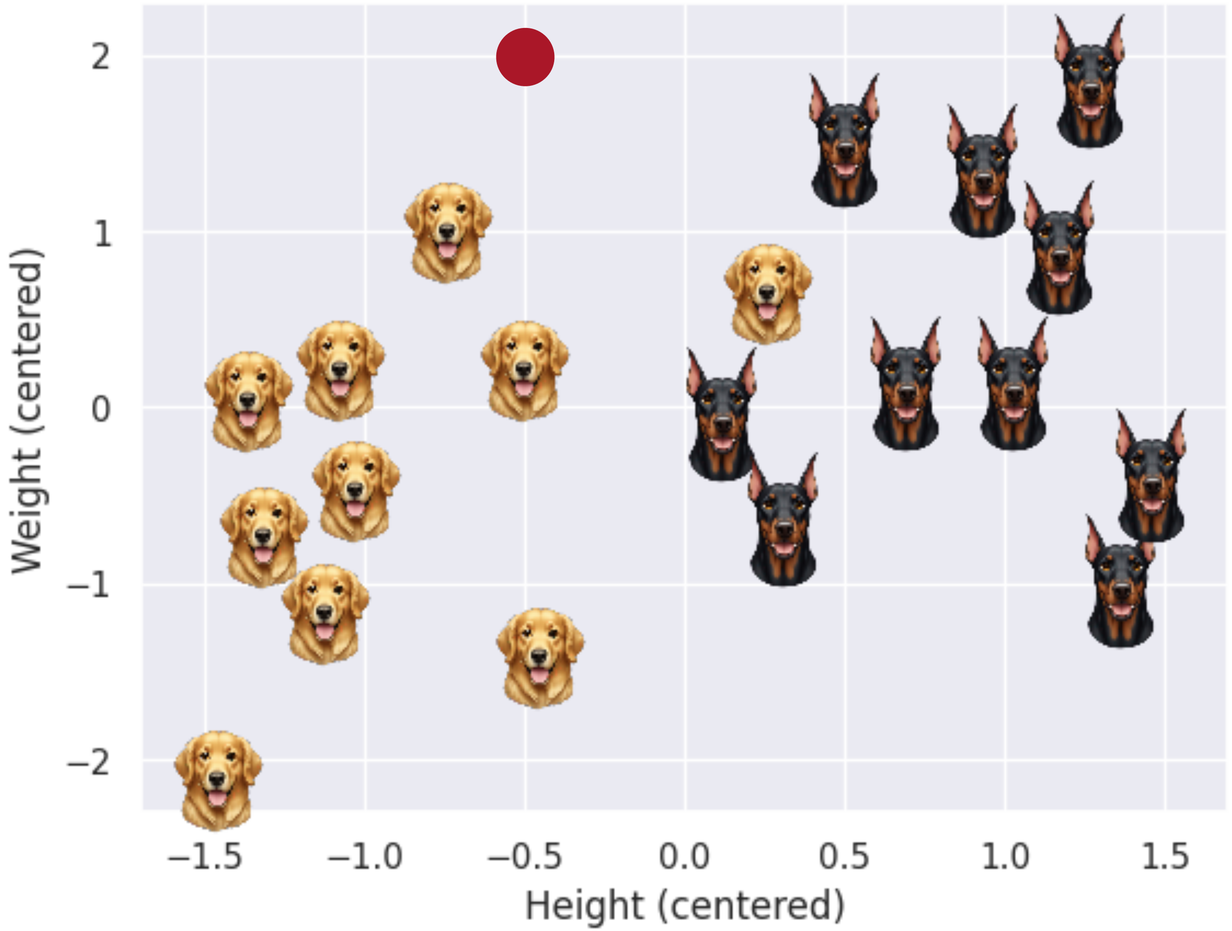
Poll

- Inputs: $x_0 = -0.5,\ x_1 = 2$
- Weights: $w_0 = -2,\ w_1 = -0.1$
- Compute weighted sum
  $S = w_0 x_0 + w_1 x_1$
- Output
  $\text{sign}(S) = +1$ if $S > 0$
  $= -1$ if $S \leq 0$

What is the weighted sum S?

A. -1.2 C. 0.8

B. -0.2 D. 1.0

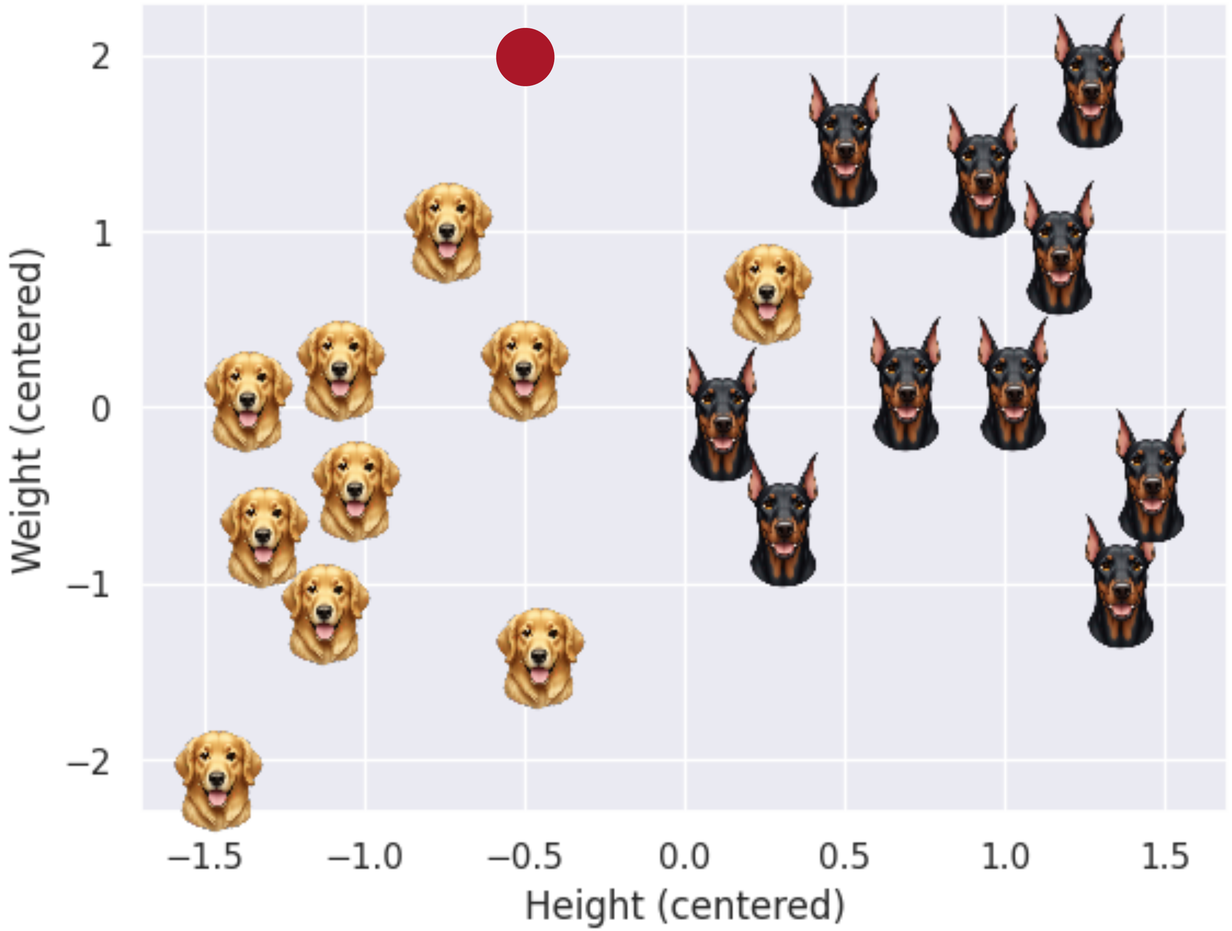


Golden retriever: +1, Doberman -1

Poll

- Inputs: $x_0 = -0.5,\ x_1 = 2$
- Weights: $w_0 = -2,\ w_1 = -0.1$
- Compute weighted sum
  $S = w_0 x_0 + w_1 x_1$
- Output
  $\text{sign}(S) = +1$ if $S > 0$
  $= -1$ if $S \leq 0$

What is the weighted sum S?



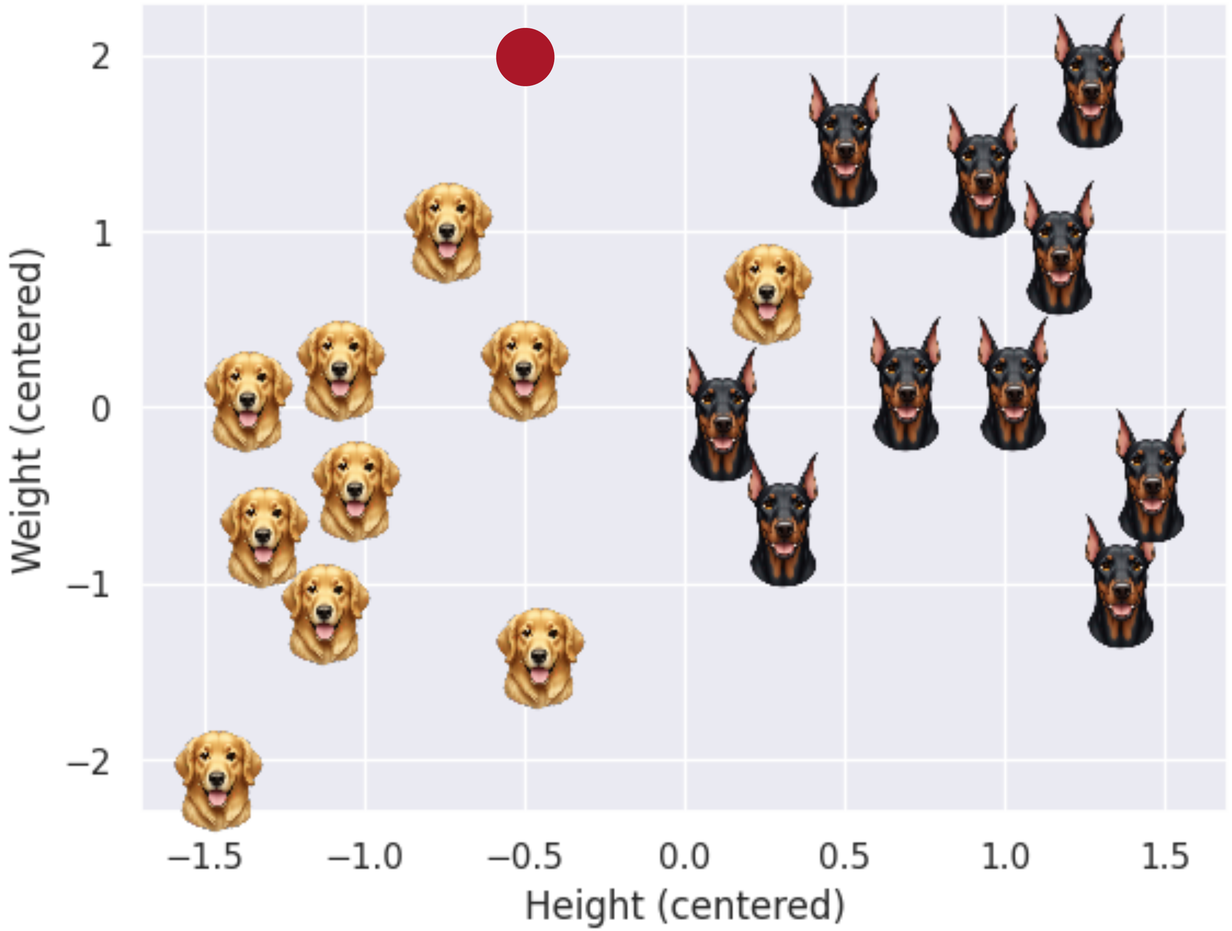
Golden retriever: +1, Doberman -1

Perceptron

- Inputs: $x_0 = -0.5,\ x_1 = 2$
- Weights: $w_0 = -2,\ w_1 = -0.1$
- Compute weighted sum
  $S = w_0 x_0 + w_1 x_1$
- Output
  $\text{sign}(S) = +1$ if $S > 0$
  $= -1$ if $S \leq 0$

$$S = w_0 x_0 + w_1 x_1$$
$$= (-2) \cdot (-0.5) + (-0.1) \cdot 2$$
$$= 1 + (-0.2) = 0.8$$

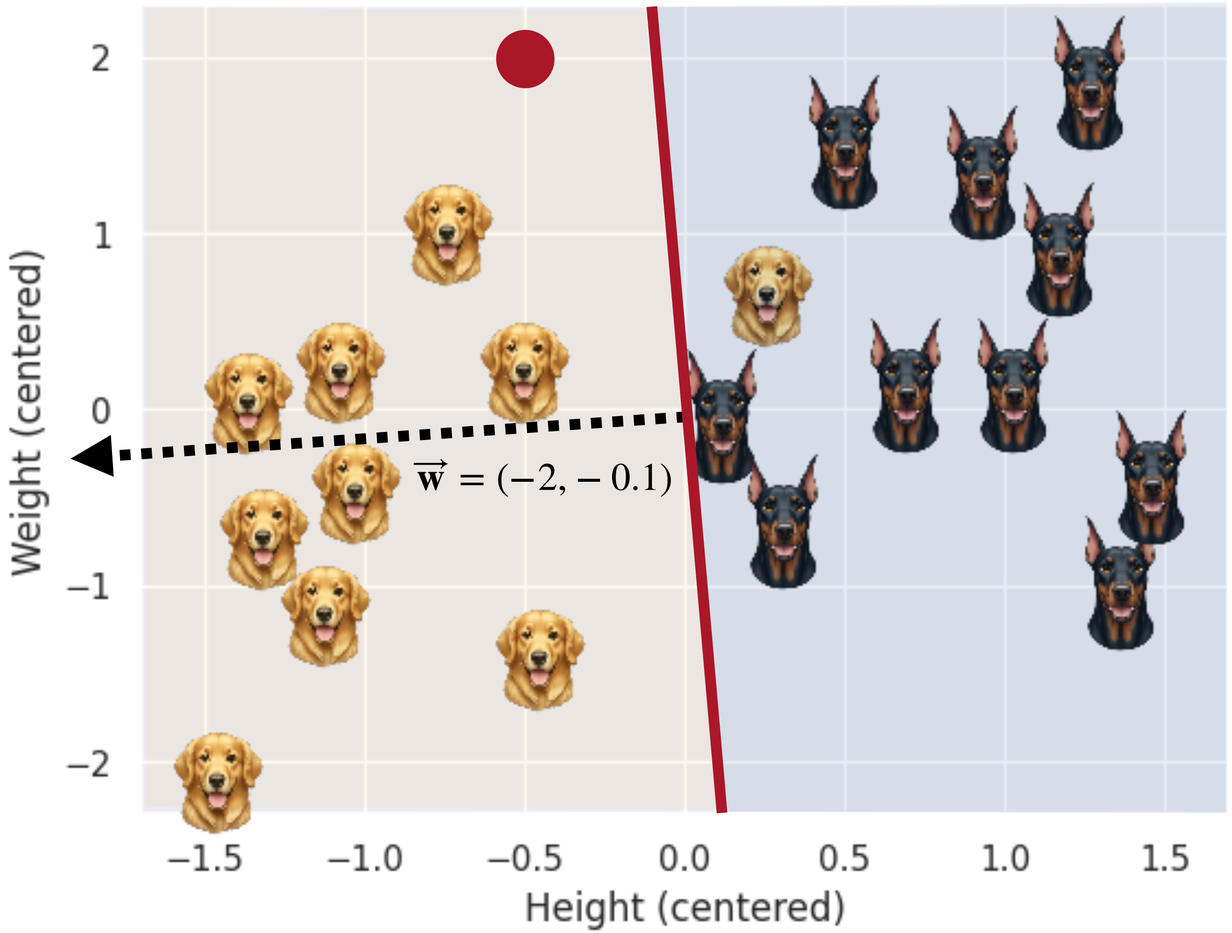
$$\text{sign}(S) = \text{sign}(0.8) = +1$$



Golden retriever: +1, Doberman -1

Perceptron: Geometric Perspective

- Inputs: $x_0 = -0.5,\ x_1 = 2$
- Weights: $w_0 = -2,\ w_1 = -0.1$
- Compute weighted sum
  $S = w_0 x_0 + w_1 x_1 = 0.8$
- Output
  $\text{sign}(S) = +1$
- Weight vector $\vec{\mathbf{w}}$ is **perpendicular** to linear decision boundary.
- Datapoints on the "side" with weight vector $\vec{\mathbf{w}}$ are predicted as $+1$; those on the "opposite site" as -1.



Golden retriever: +1, Doberman -1

Ingredients of ML

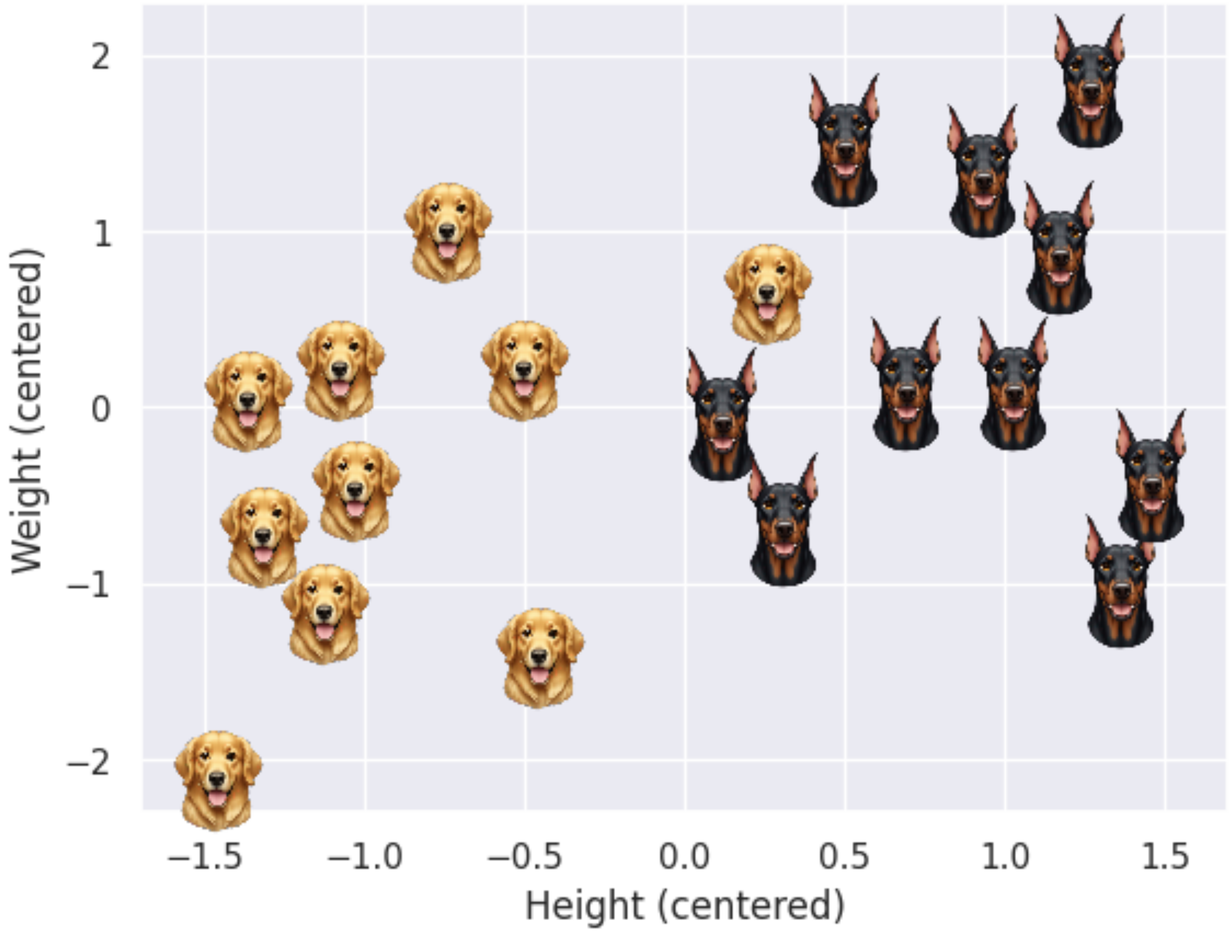
1. **Data.** Height and weight (x_0, x_1) and label $y \in \{+1, -1\}$ for Golden Retriever or Doberman.
2. **Model.** Perceptron model.
3. **Model parameters**. Weight vector $\vec{\mathbf{w}} = (w_0, w_1)$.
4. **Training.** Given datapoints, find good model parameters.
5. **Testing.** Evaluate the performance of learned model on new, previously unseen data (i.e. test set).

Perceptron: Training

Initialize all weights to 0.

For each training example with label $y \in \{+1, -1\}$:

- Compute predicted output $\hat{y} = \text{sign}(S)$
- Update weights if incorrect (i.e. $\hat{y} \neq y$)



Golden retriever: +1, Doberman -1

Perceptron: Update Rule

For each training example with label $y \in \{+1, -1\}$:

- Compute predicted output

  $S = w_0 x_0 + w_1 x_1$

  $\hat{y} = \text{sign}(S) = +1 \text{ if } S > 0$

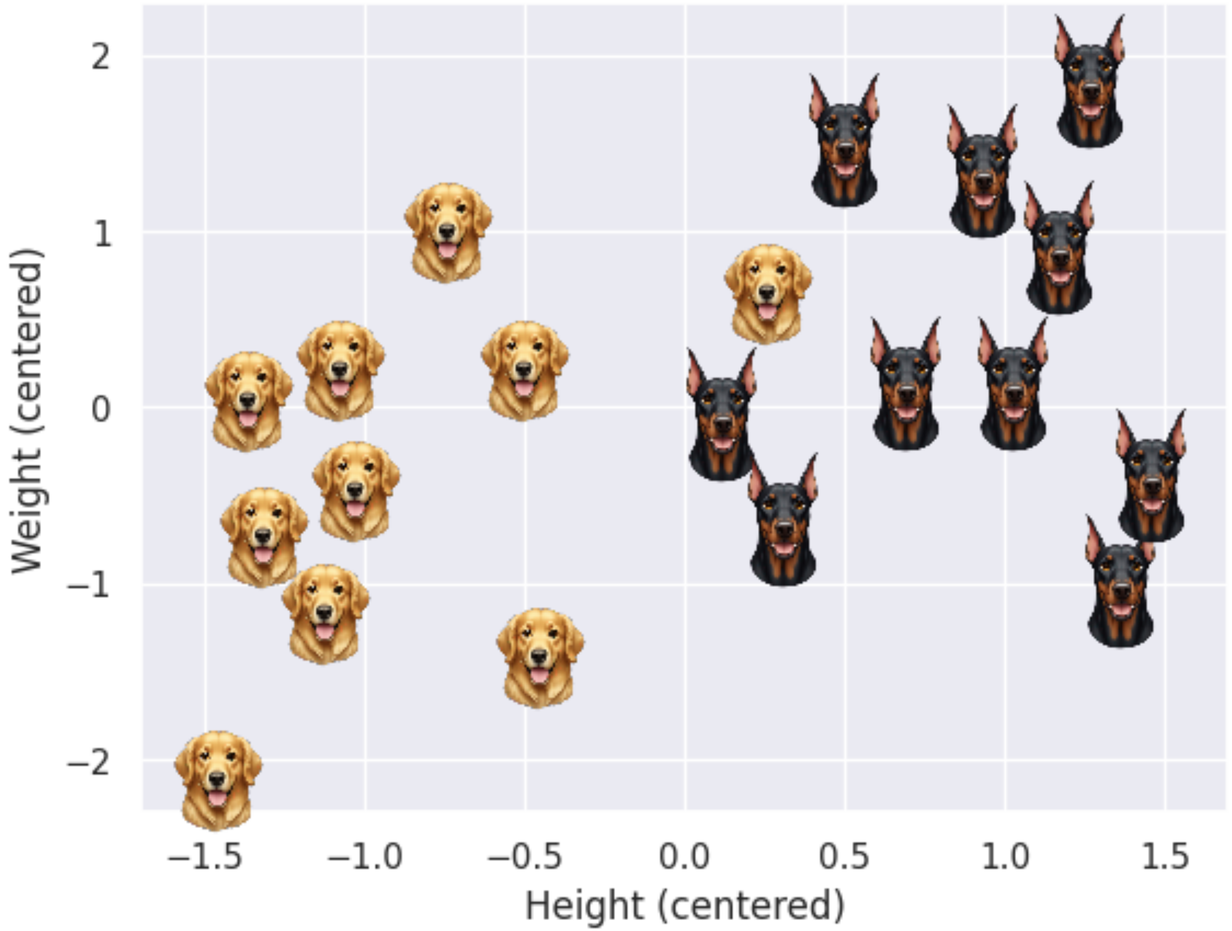
  $= -1 \text{ if } S \leq 0$

- If correct ($y = \hat{y}$), do nothing.
- If false positive ($y = -1, \hat{y} = +1$):

  $w_j' = w_j - x_j$

- If false negative ($y = +1, \hat{y} = -1$):

  $w_j' = w_j + x_j$



Golden retriever: +1, Doberman -1

Poll

- Inputs: $x_0 = 0.3,\ x_1 = 0.6,\ y = +1$
- Weights: $w_0 = -2,\ w_1 = -0.1$
- Compute predicted output

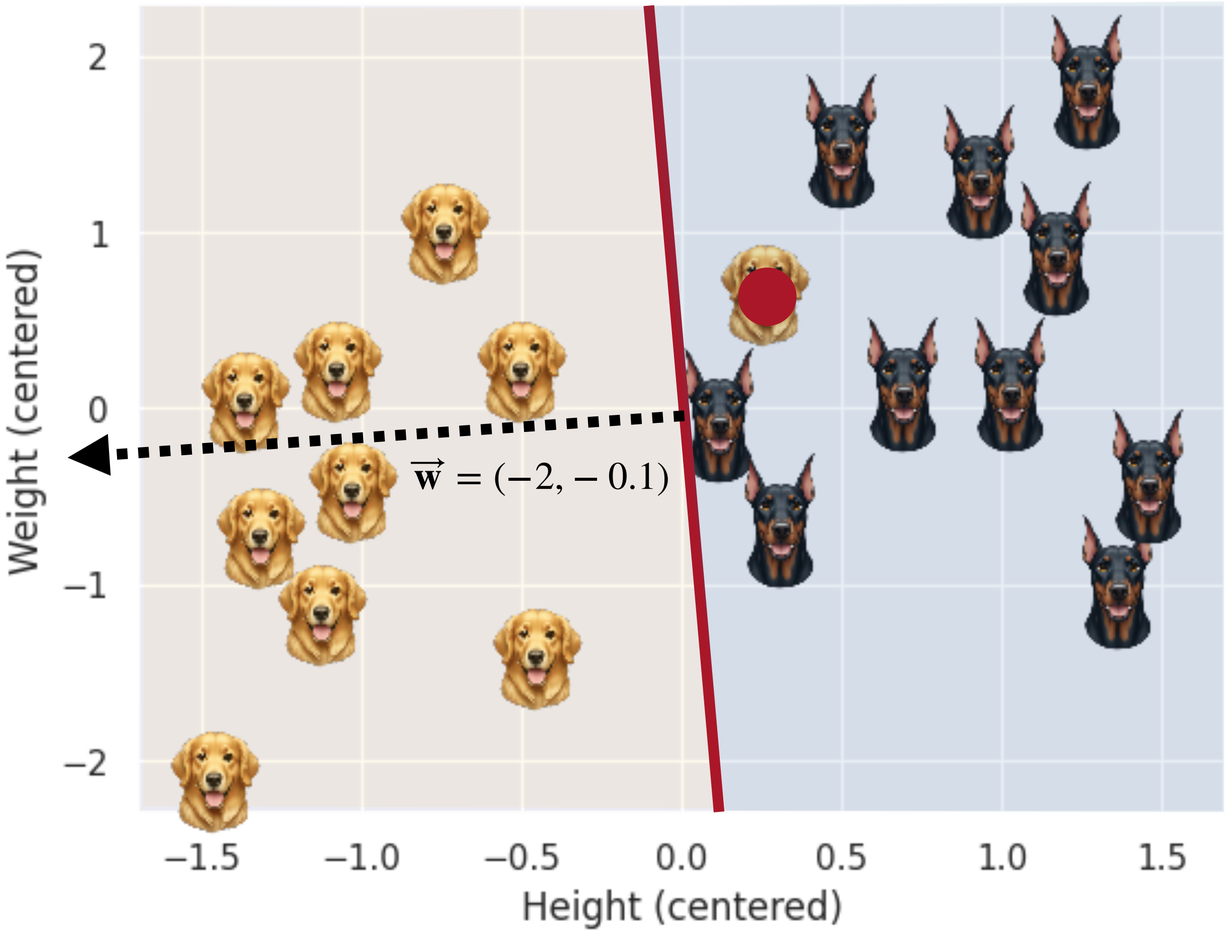
$$S = w_0 x_0 + w_1 x_1$$
$$\hat{y} = \text{sign}(S) = +1 \text{ if } S > 0$$
$$= -1 \text{ if } S \leq 0$$

Which scenario is true?

A. Correct prediction ($y = \hat{y}$)

B. False positive ($y = -1,\ \hat{y} = +1$)

C. False negative ($y = +1,\ \hat{y} = -1$)



Golden retriever: +1, Doberman -1

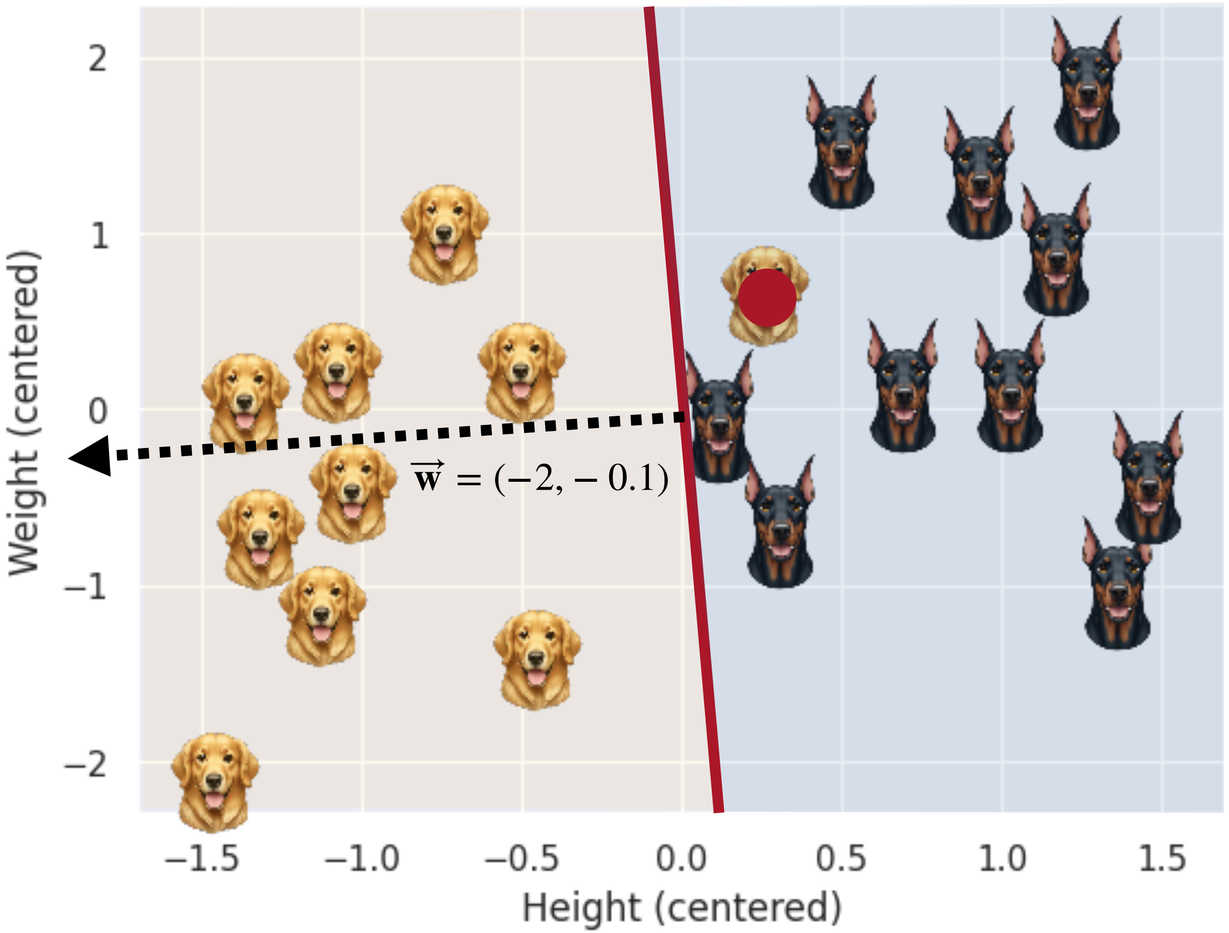
Poll

- Inputs: $x_0 = 0.3,\ x_1 = 0.6,\ y = +1$
- Weights: $w_0 = -2,\ w_1 = -0.1$

$$
\begin{aligned}
S &= w_0 x_0 + w_1 x_1 \\
&= (-2) \cdot 0.3 + (-0.1) \cdot 0.6 \\
&= -0.6 + (-0.06) = -0.66
\end{aligned}
$$

$$\hat{y} = \text{sign}(S) = -1$$

Which scenario is true?



Golden retriever: +1, Doberman -1

Perceptron: Update Rule

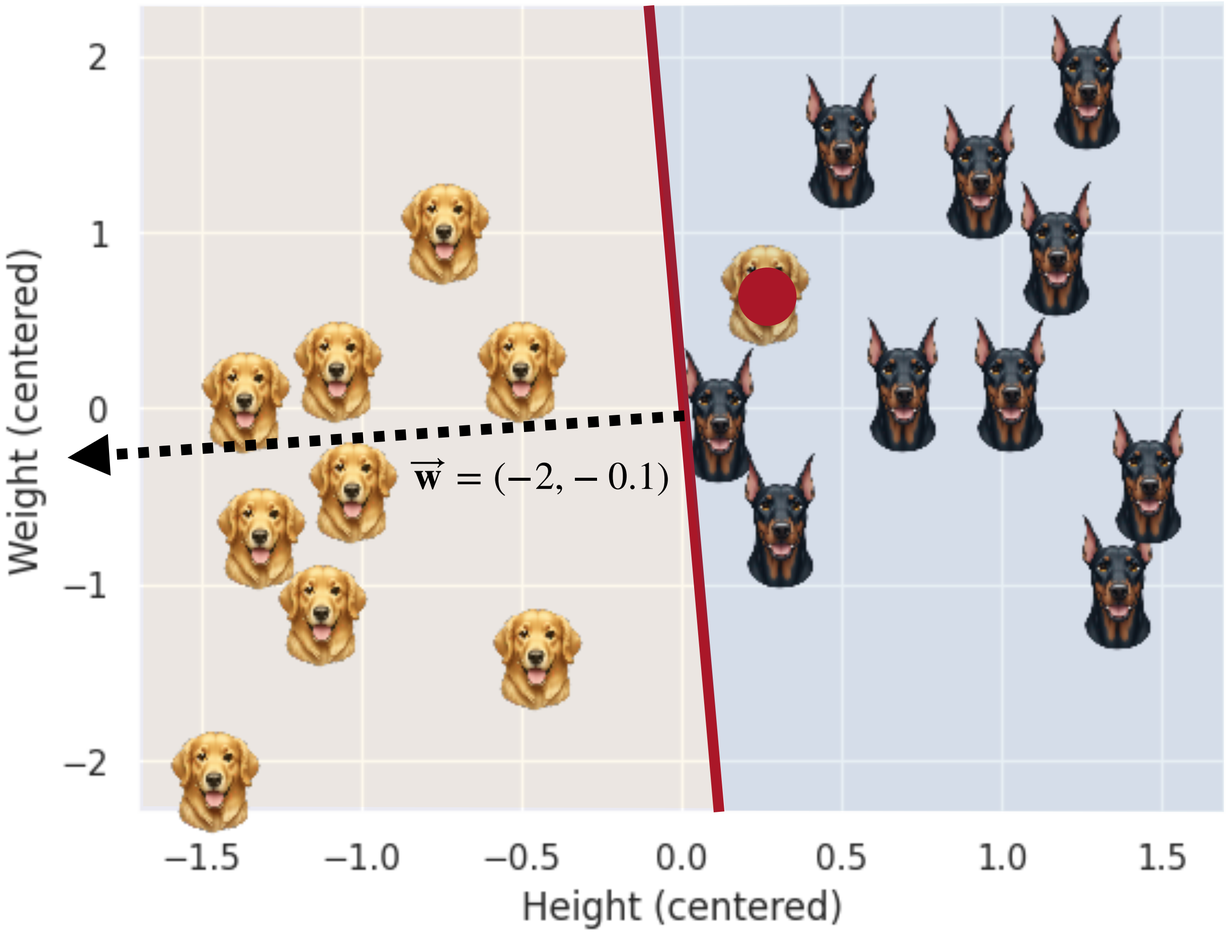
- Inputs: $x_0 = 0.3,\ x_1 = 0.6,\ y = +1$
- Weights: $w_0 = -2,\ w_1 = -0.1$

$$S = w_0 x_0 + w_1 x_1 = -0.66$$

$$\hat{y} = \text{sign}(S) = -1$$

Update rules:

- If correct ($y = \hat{y}$), do nothing.
- If false positive ($y = -1,\ \hat{y} = +1$):
  $w_j' = w_j - x_j$
- If false negative ($y = +1,\ \hat{y} = -1$):
  $w_j' = w_j + x_j$



Perceptron: Update Rule

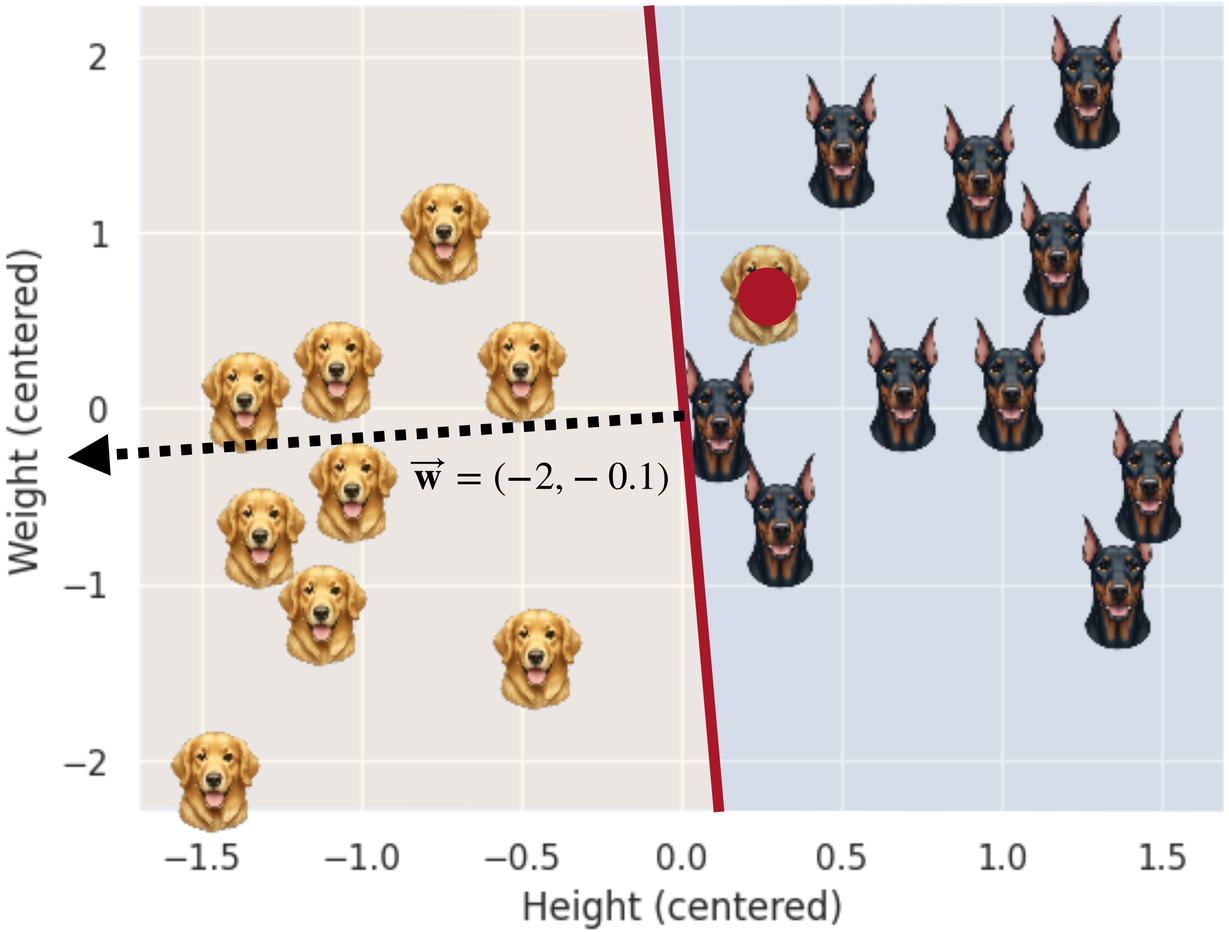
- Inputs: $x_0 = 0.3,\ x_1 = 0.6,\ y = +1$
- Weights: $w_0 = -2,\ w_1 = -0.1$

$$S = w_0 x_0 + w_1 x_1 = -0.66$$

$$\hat{y} = \text{sign}(S) = -1$$

Update rules:

- If correct ($y = \hat{y}$), do nothing.
- If false positive ($y = -1,\ \hat{y} = +1$):
  $w_j' = w_j - x_j$
- **If false negative ($y = +1,\ \hat{y} = -1$):**
  $w_j' = w_j + x_j$



Perceptron: Update Rule

- Inputs: $x_0 = 0.3,\ x_1 = 0.6,\ y = +1$
- Weights: $w_0 = -2,\ w_1 = -0.1$

$$S = w_0 x_0 + w_1 x_1 = -0.66$$

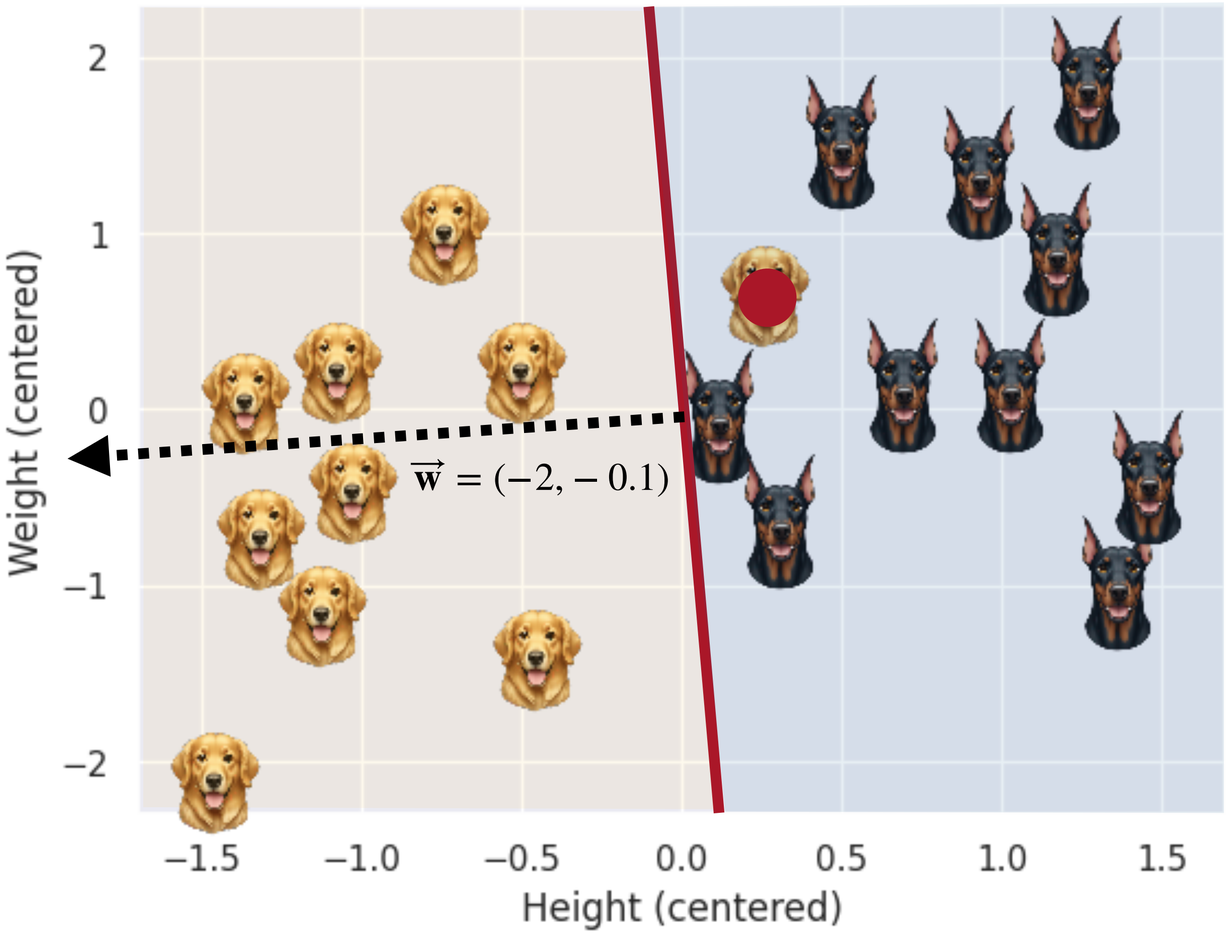
$$\hat{y} = \text{sign}(S) = -1$$

If false negative ($y = +1,\ \hat{y} = -1$):

$$w_j' = w_j + x_j$$

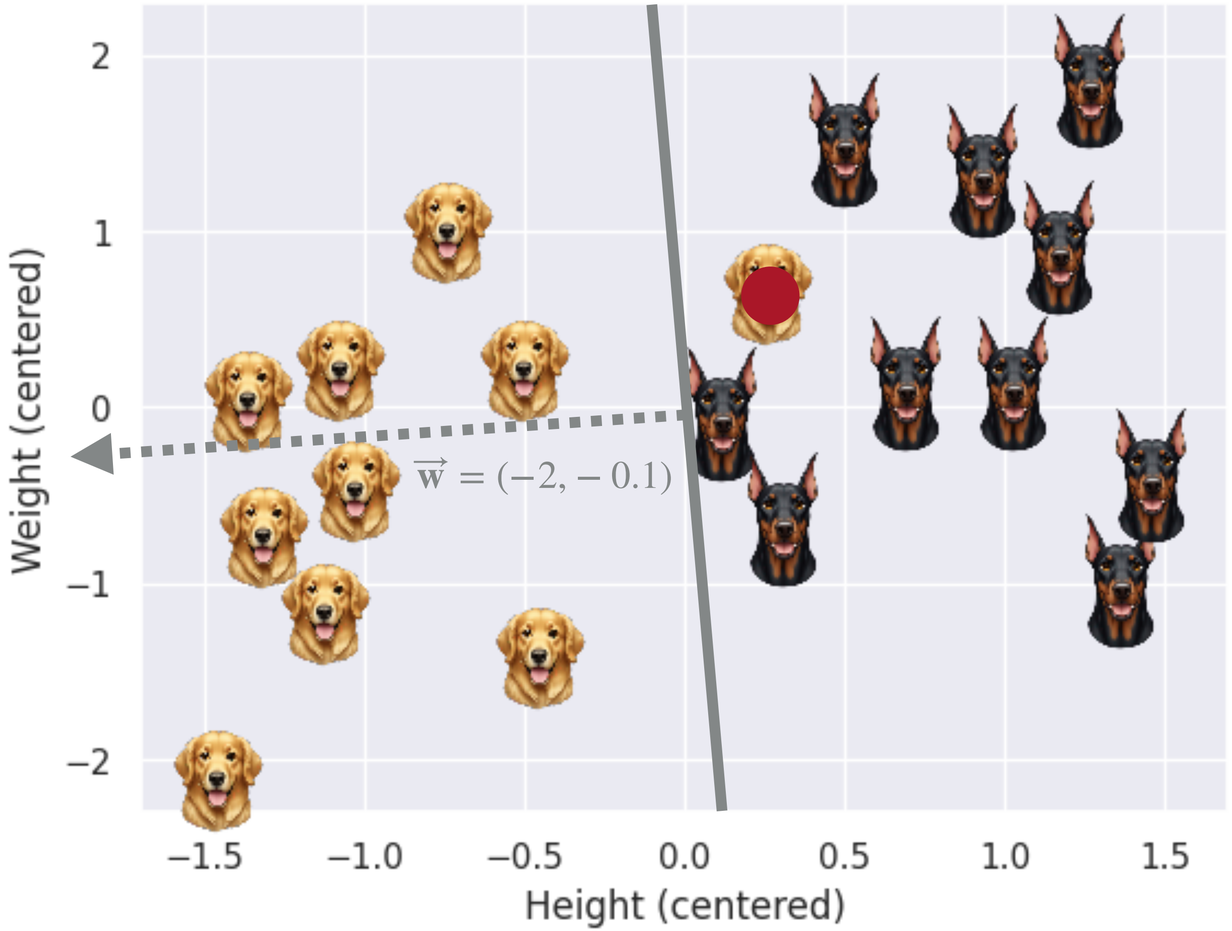
$$w_0' = w_0 + x_0 = -2 + 0.3 = -1.7$$

$$w_1' = w_1 + x_1 = -0.1 + 0.6 = 0.5$$



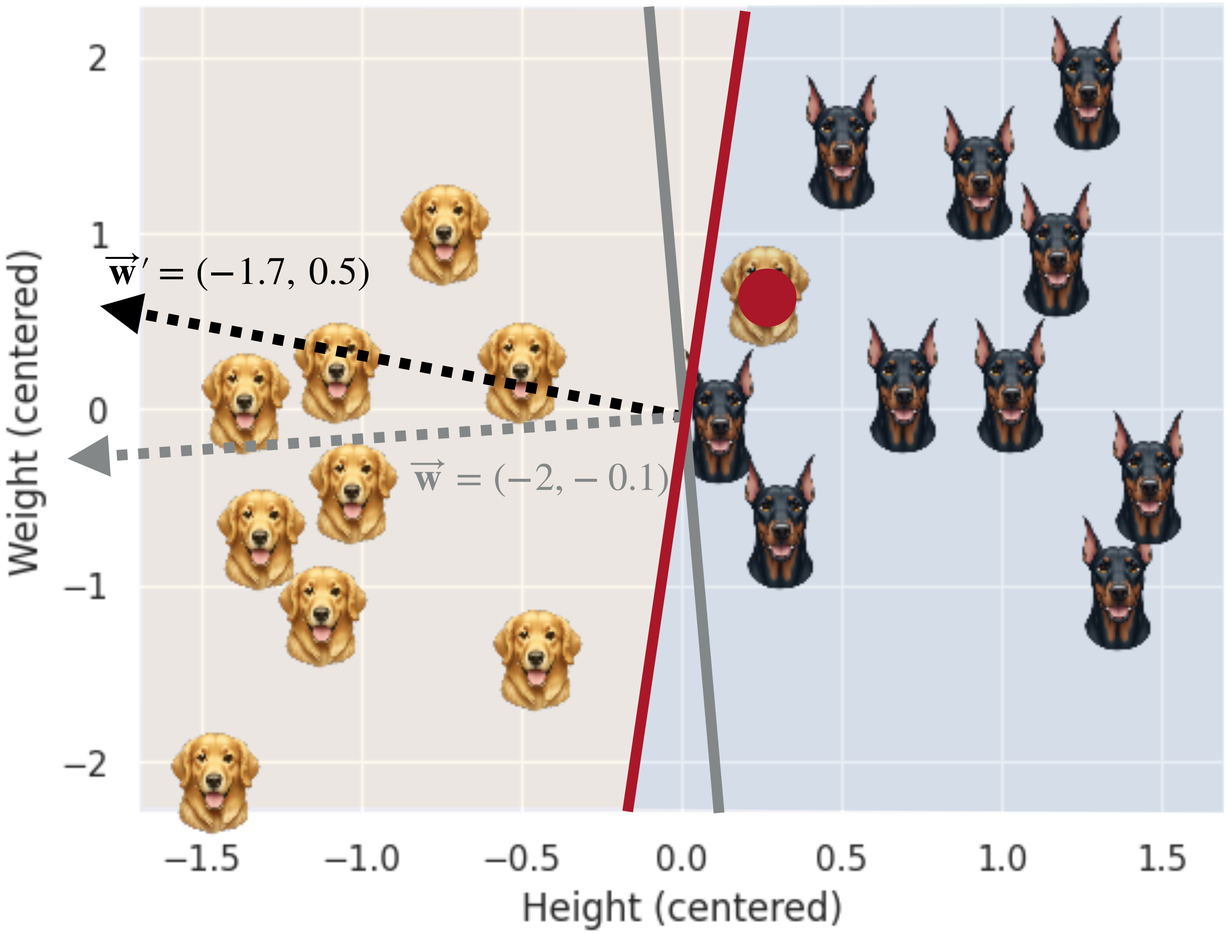
Perceptron: Update Rule

- Inputs: $x_0 = 0.3,\ x_1 = 0.6,\ y = +1$
- Old weights: $w_0 = -2,\ w_1 = -0.1$



Perceptron: Update Rule

- Inputs: $x_0 = 0.3,\ x_1 = 0.6,\ y = +1$
- Old weights: $w_0 = -2,\ w_1 = -0.1$
- New weights: $w_0' = -1.7,\ w_1' = 0.5$



Perceptron: Update Rule

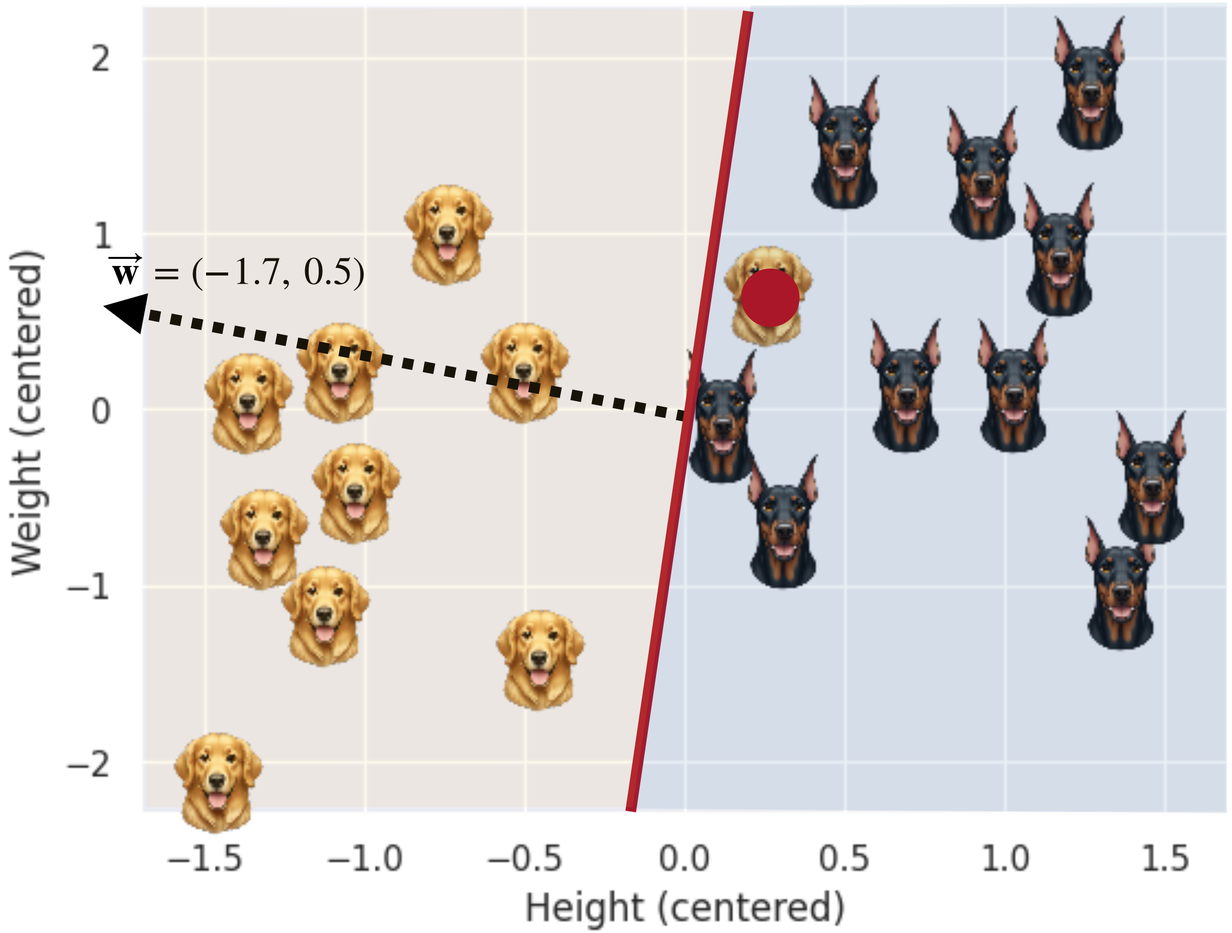
- Inputs: $x_0 = 0.3,\ x_1 = 0.6,\ y = +1$
- New weights: $w_0 = -1.7,\ w_1 = 0.5$

$$S = w_0 x_0 + w_1 x_1 = -0.21$$

$$\hat{y} = \text{sign}(S) = -1$$

Still false negative, but closer to boundary.

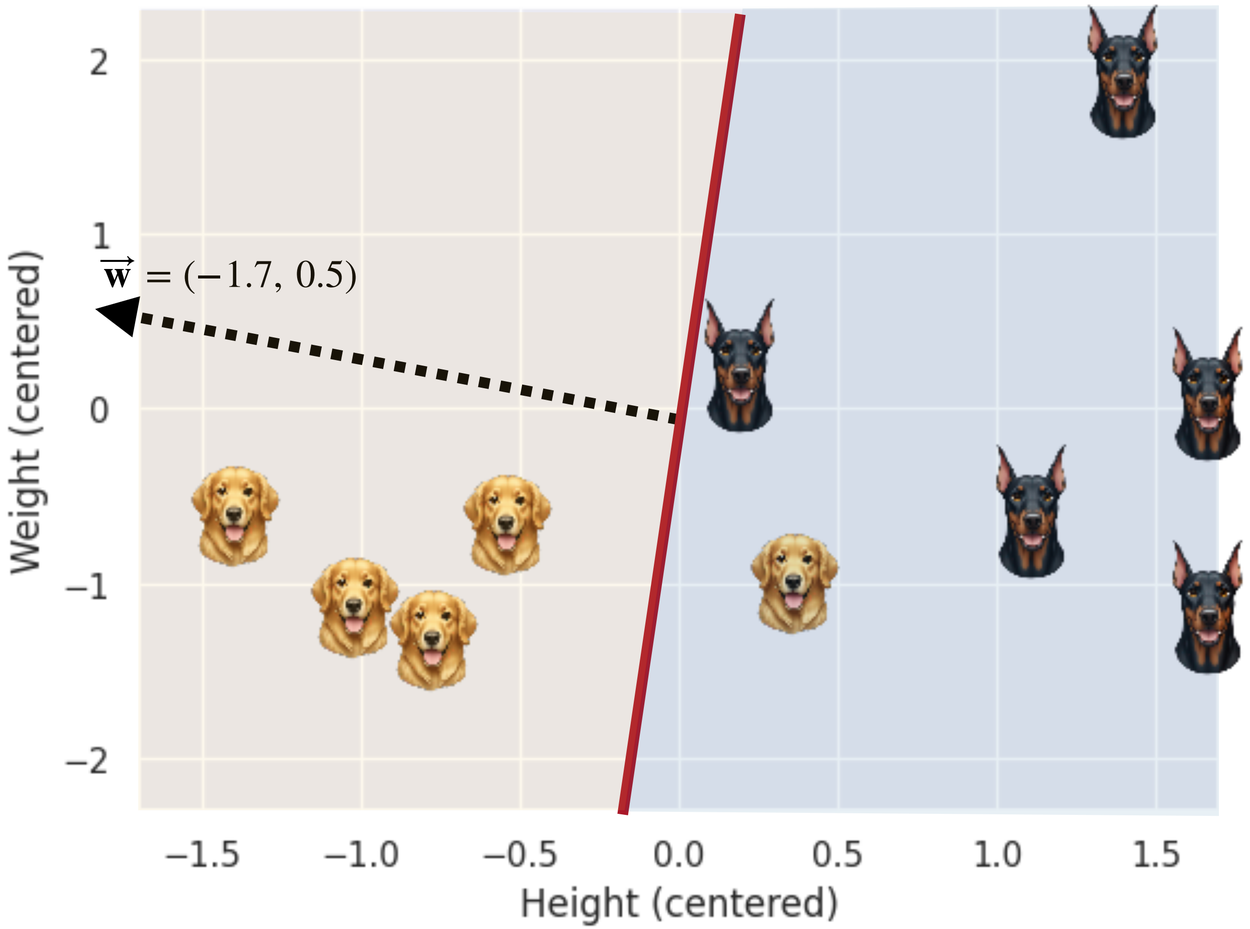
Intuition: Update rule moves decision boundary to be better for the current training example.



Perceptron: Testing

Evaluate performance using an evaluation metric on an unseen set of examples (e.g. test examples).

$$\text{Error rate} = \frac{\text{\# incorrect}}{\text{\# total}}$$





Poll

Evaluate performance using an evaluation metric on an unseen set of examples (e.g. test examples).

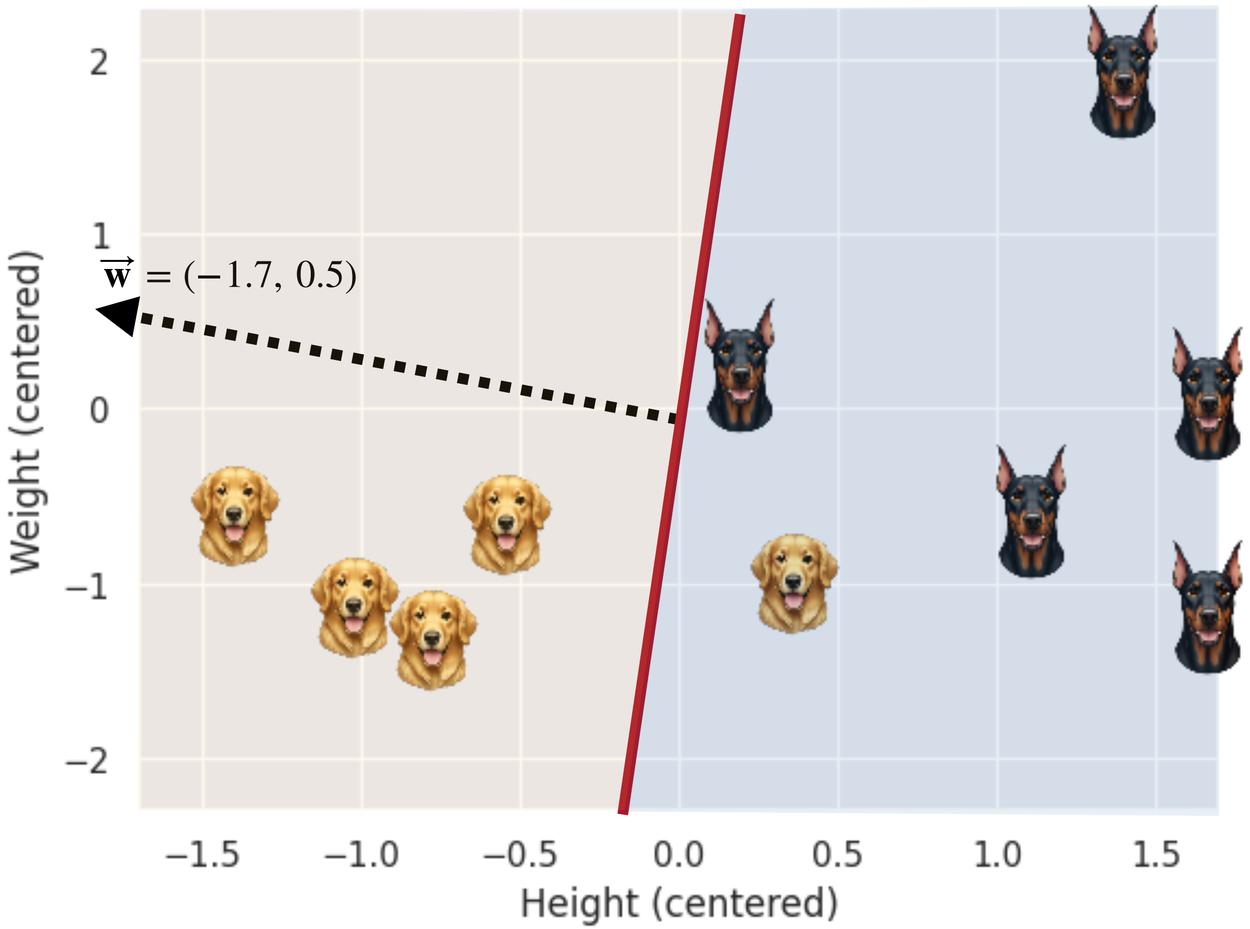
$$\text{Error rate} = \frac{\text{\# incorrect}}{\text{\# total}}$$

Consider the following unseen examples (5 Golden Retrievers, 5 Dobermans) and the given Perceptron model.

What is the error rate on test set?

A. $0/10$ C. $2/10$

B. $1/10$ D. $9/10$





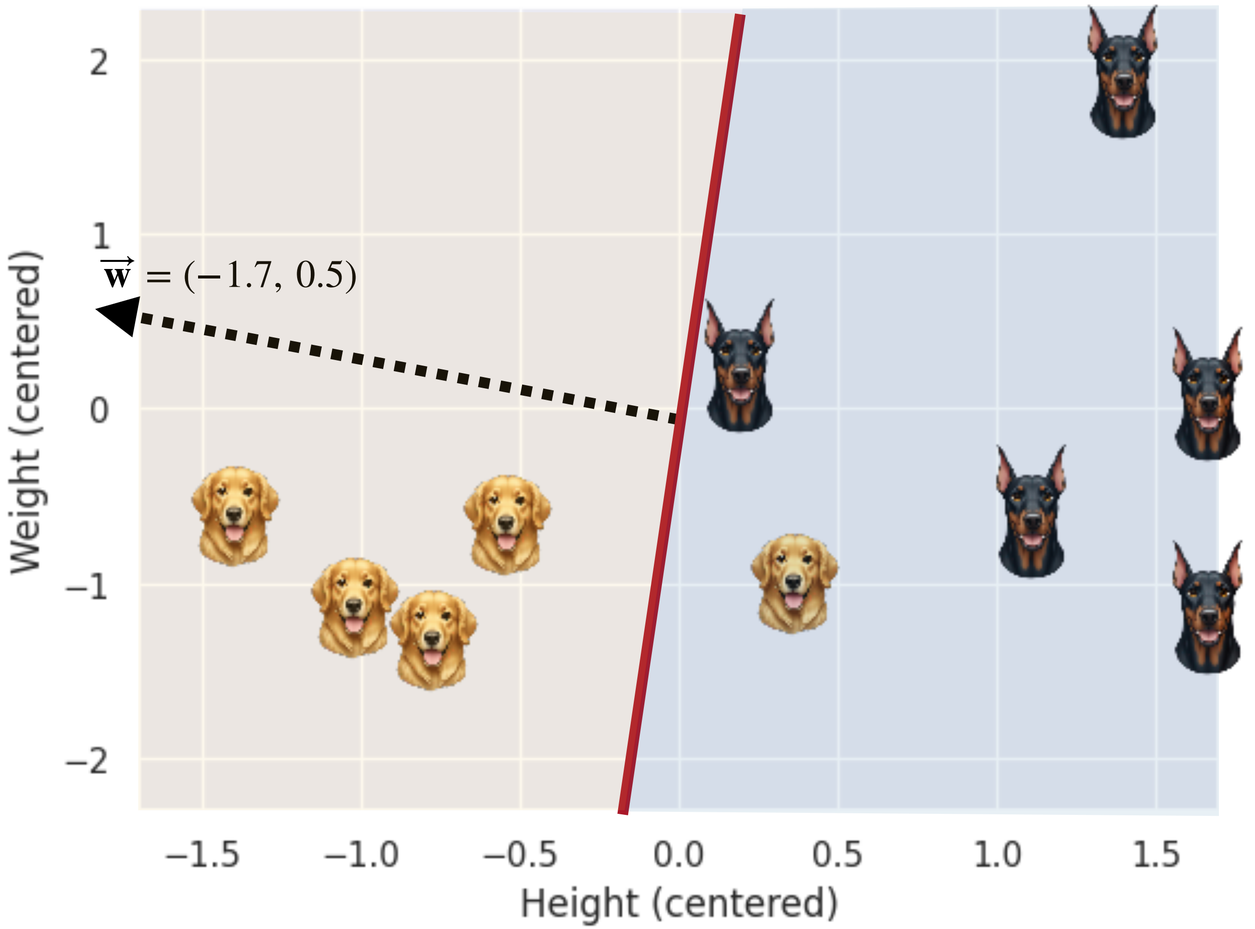
Poll

Evaluate performance using an evaluation metric on an unseen set of examples (e.g. test examples).

$$\text{Error rate} = \frac{\text{\# incorrect}}{\text{\# total}}$$

Consider the following unseen examples (5 Golden Retrievers, 5 Dobermans) and the given Perceptron model.

What is the error rate on test set?

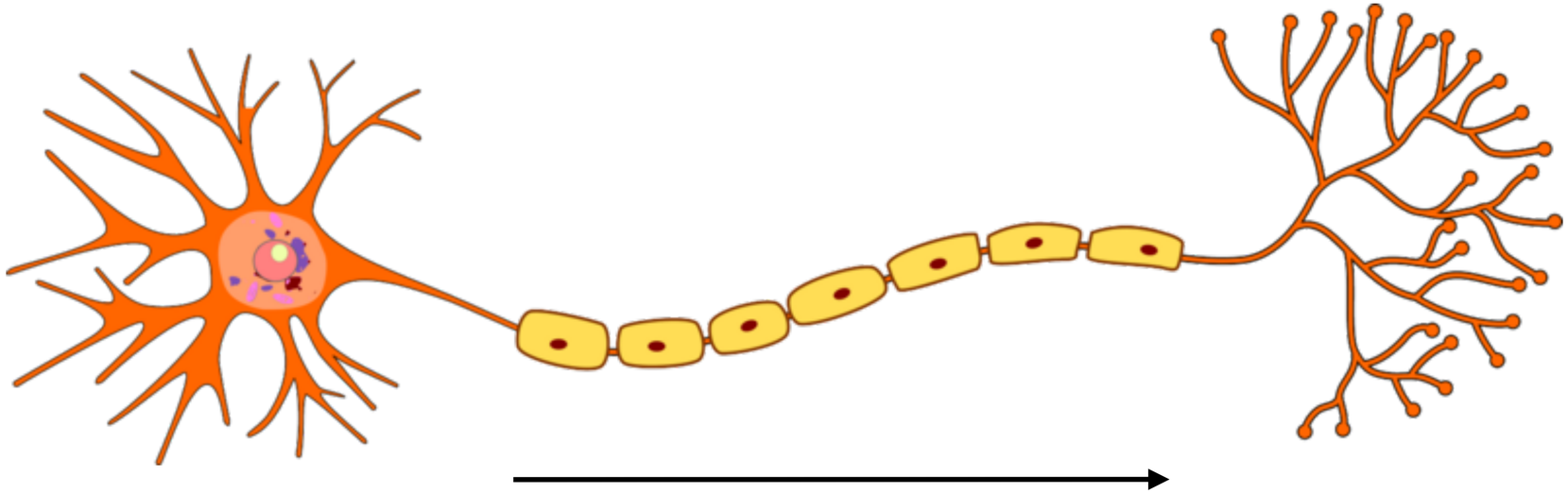


Ingredients of ML

1. **Data.** Height and weight (x_0, x_1) and label $y \in \{+1, -1\}$ for Golden Retriever or Doberman.
2. **Model.** Perceptron model.
3. **Model parameters**. Weight vector $\overrightarrow{\mathbf{w}} = (w_0, w_1)$.
4. **Training.** Use the Perceptron update rule to train on training examples.
5. **Testing.** Compute error rate on new, previously unseen data (i.e. test set).

Multi-layer Perceptron

Biological Neuron



Dendrites

Receive information from other neurons

Action potential

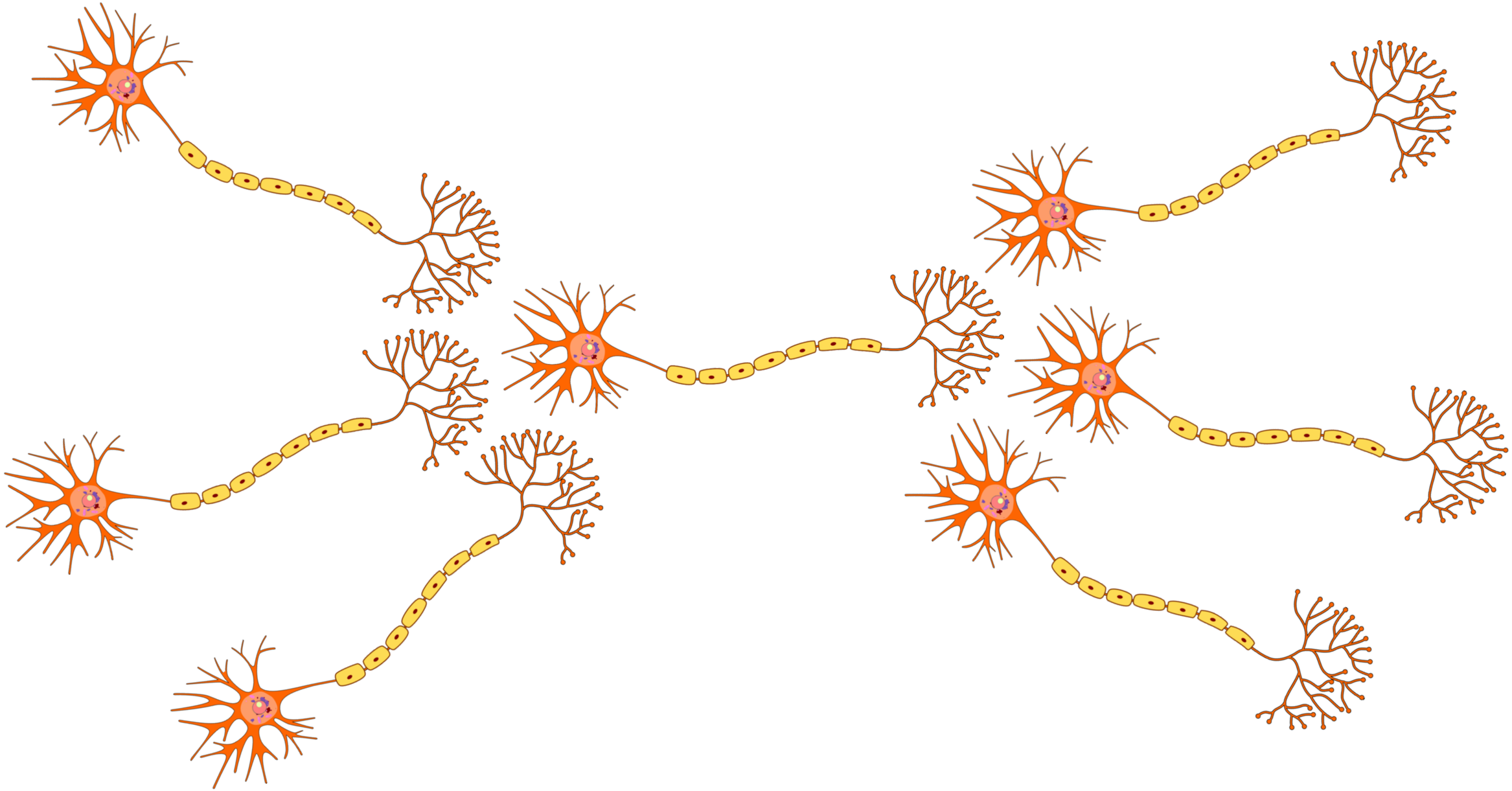
Neuron “fires” based on received information

Synaptic terminals

Relay “firing” to other neurons

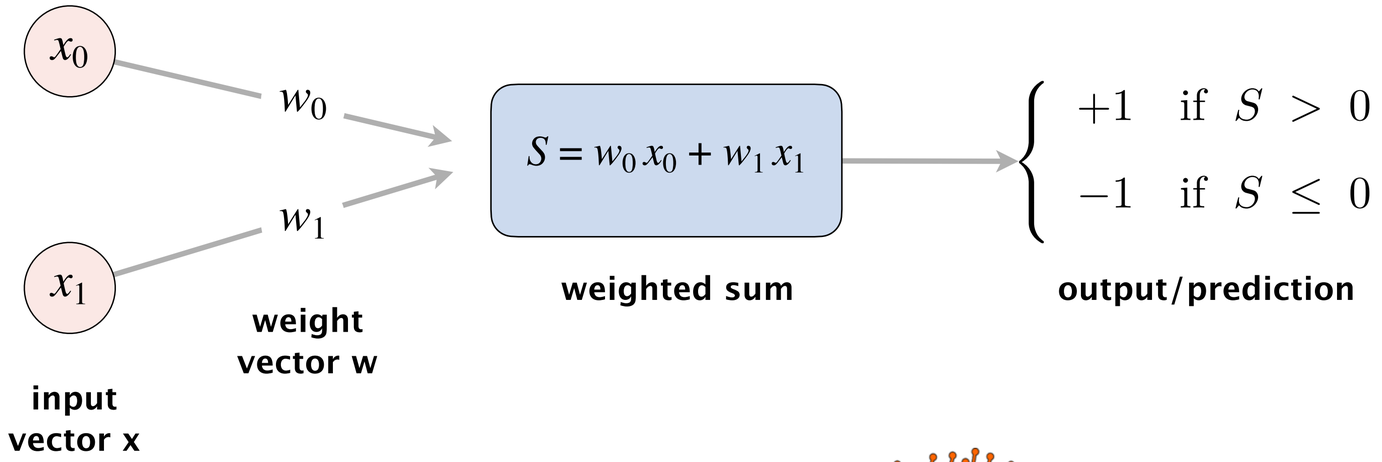
Neuron firing (or not) is a binary output.

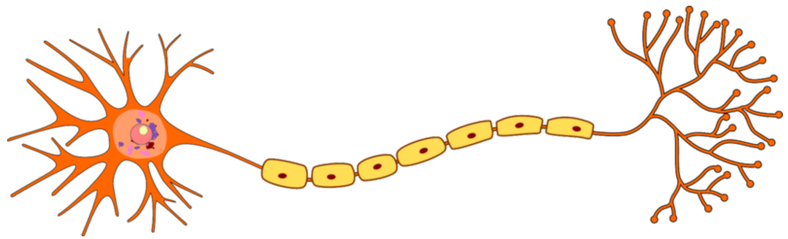
Biological Neuron



Perceptron

Takes several inputs and produces a single binary output.





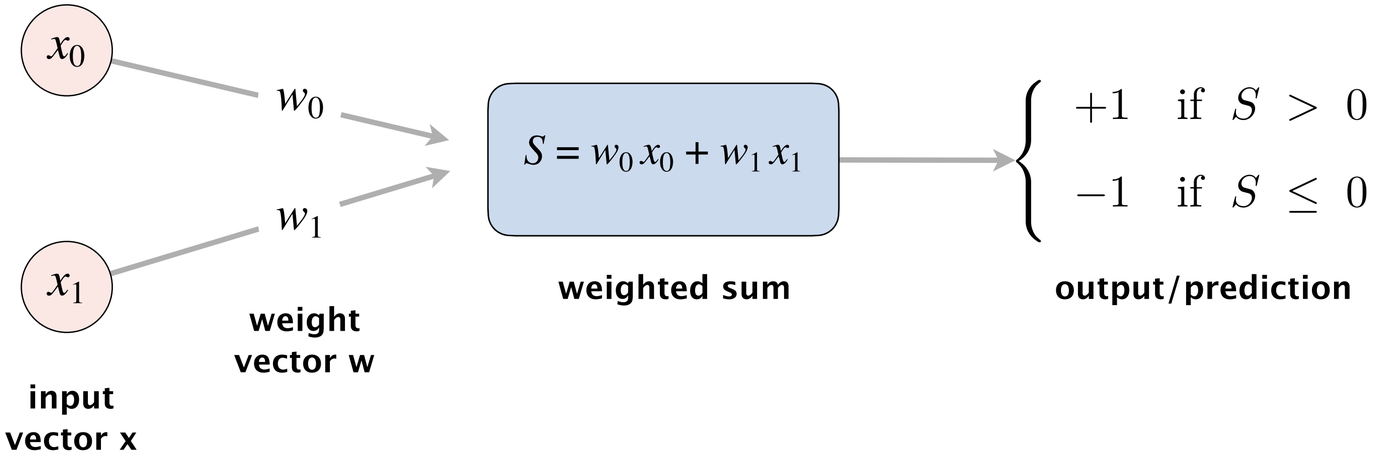
Receive input

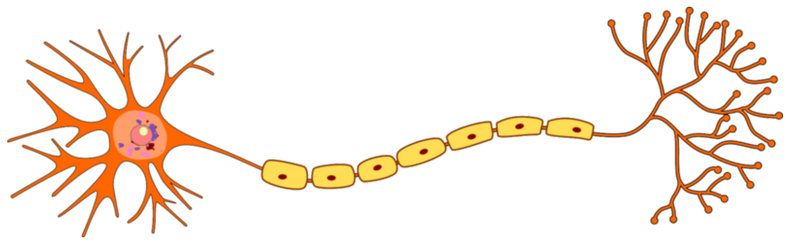
Binary "firing"

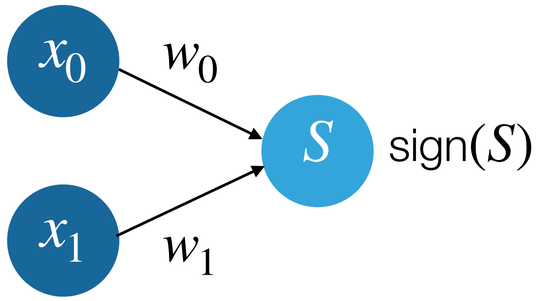
Relay output

Perceptron

Takes several inputs and produces a single binary output.



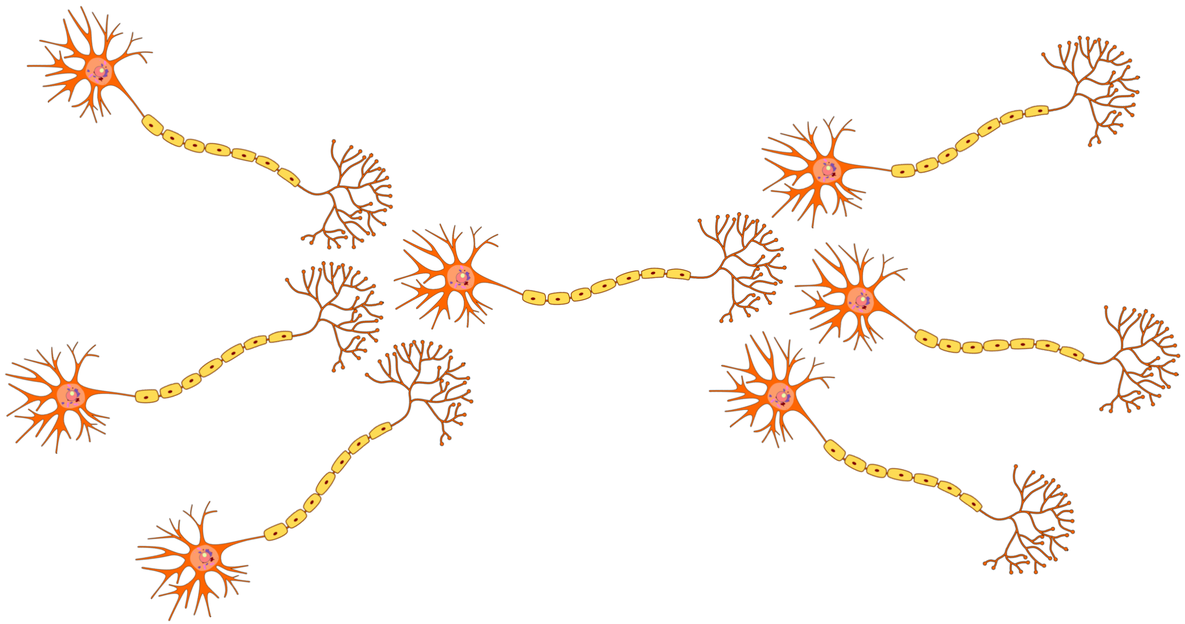


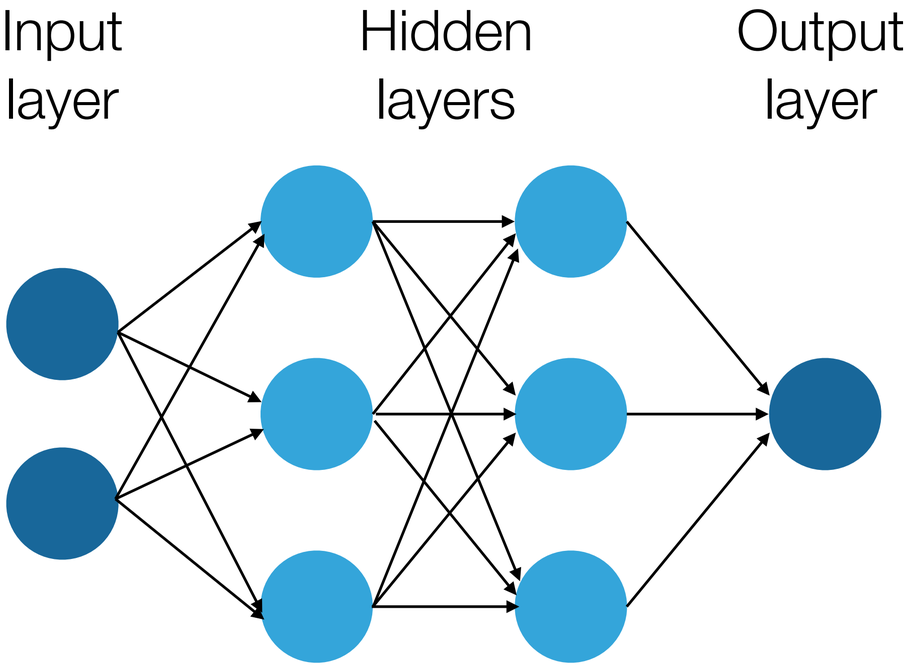


Multi-Layer Perceptron (MLP)

Composed of multiple layers of perceptron neurons, where each neuron is connected to all neurons in the next layer (i.e. fully-connected).

MLPs are an example of a deep neural network.



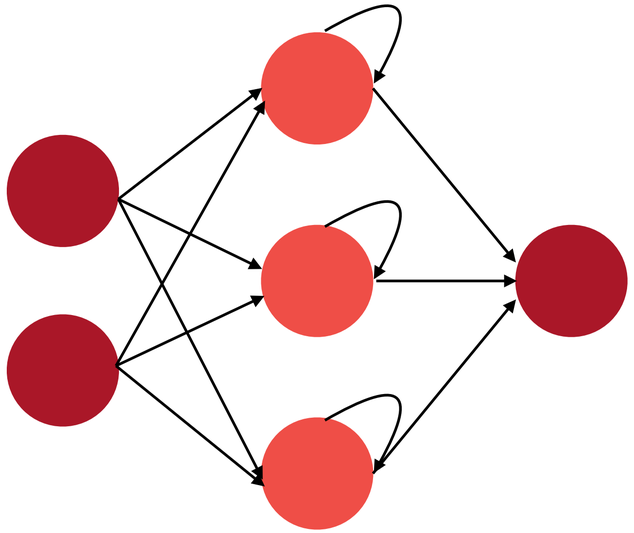


Deep Learning

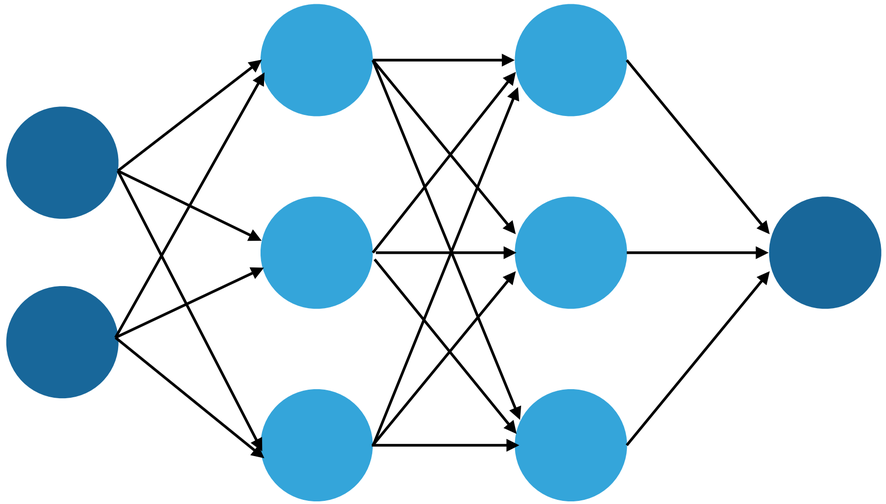
Refers to ML models that use deep neural networks.

Popular neural network architectures:

- Recurrent neural networks
- Convolutional neural networks
- Transformers



Recurrent NN



Fully-connected NN

Multi-class Perceptron

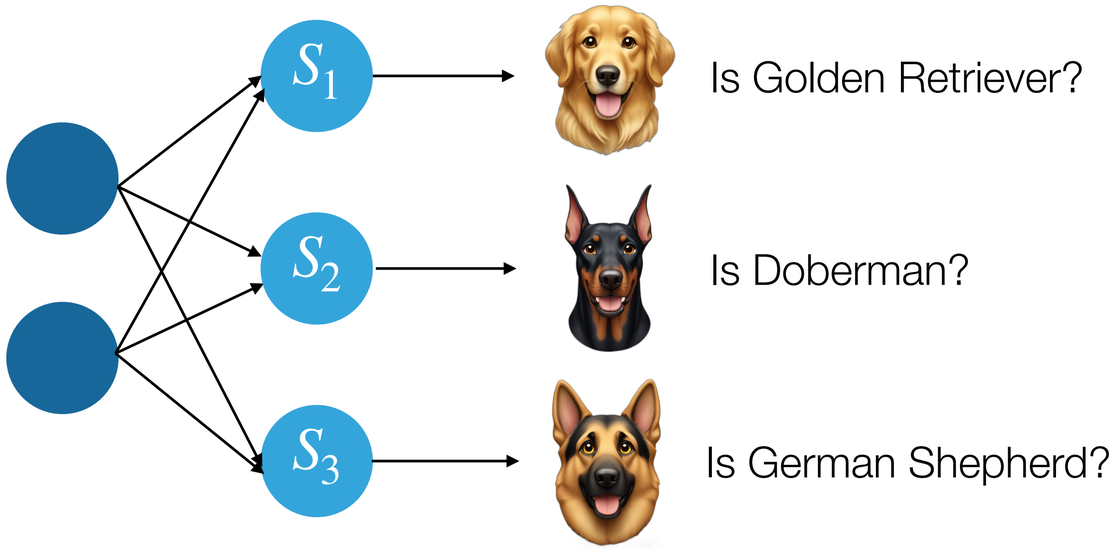
Multi-class Classification

Predict one of **k** classes (k > 2).

Suppose we have **k = 3** dog species we are trying to classify.

Train **k = 3** binary Perceptron models and compute **3** weighted sums: S_1, S_2, S_3.

Final prediction is the class with the largest weighted sum.



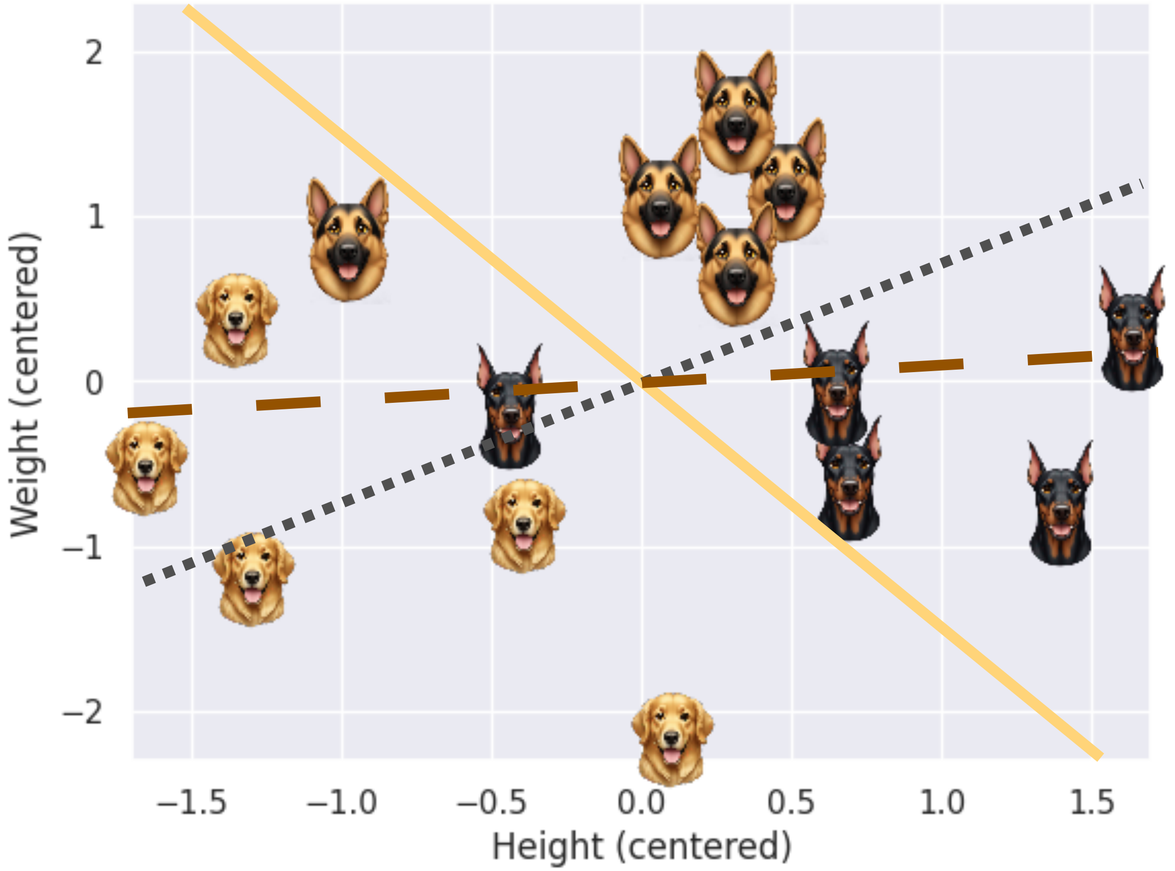
Multi-class Classification: Geometric Perspective

Predict one of **k** classes (k > 2).

Suppose we have **k = 3** dog species we are trying to classify.

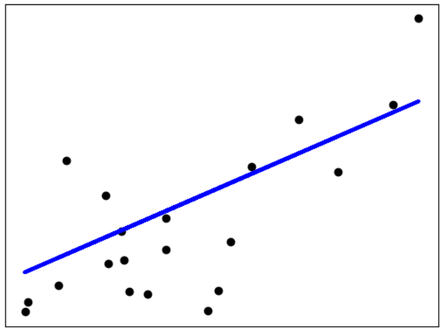
Train **k = 3** binary Perceptron models and compute **3** weighted sums: S_1, S_2, S_3.

Final prediction is the class with the largest weighted sum.



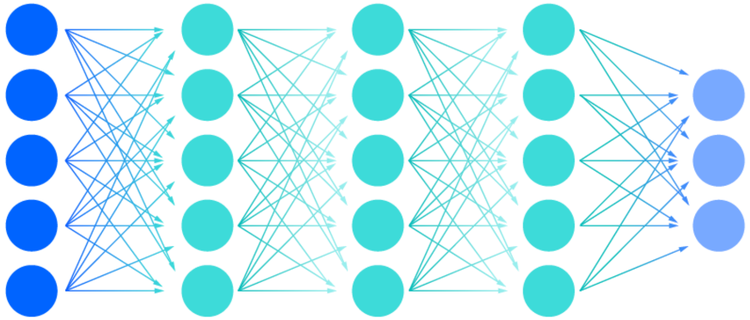
Machine learning models

ML model: an algorithm used to perform a task.

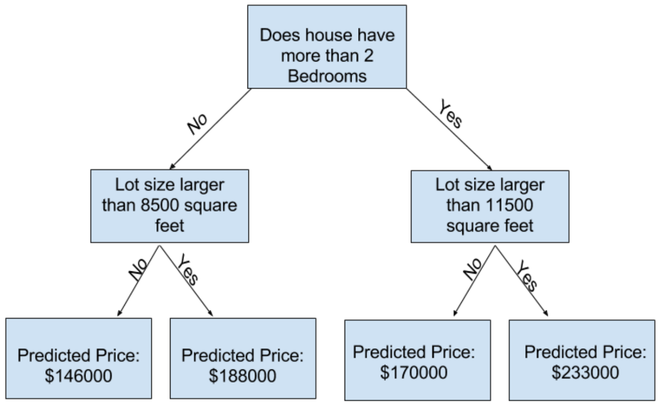


Supervised ML models:

- Linear models (e.g. perceptron)
- Deep neural networks* (e.g. multi-layer perceptron)
- Decision trees
- Ensembles (e.g. random forests)
- …



*also for unsupervised and reinforcement learning



Readings

Required:

- Etienne Bernard, Wolfram. Introduction to Machine Learning: Machine Learning Paradigms. https://www.wolfram.com/language/introduction-machine-learning/machine-learning-paradigms/

Credits

Lecture contents based in part on the following:

- COS324 instructors. COS324 lecture slides.
- Etienne Bernard, Wolfram. Introduction to Machine Learning: Machine Learning Paradigms. https://www.wolfram.com/language/introduction-machine-learning/machine-learning-paradigms/
- Shree Nayar. Perceptron | Neural Networks. https://youtu.be/OFbnpY_k7js?si=wrWOvNXE3UVCq-7Z

Credits

Media	Source	License
AlphaFold	AlphaFold	CC BY 4.0
ChatGPT screenshot	Mashable	
Cicero	Science	
DALL-E 2	Open AI	
Dancing Robot	Disney Research	
Basketball Tabular Data	Statology	
Object detection NYC	Ilija Mihajlovic	
Chat GPT market	MarketWatch	
Robotic arms	Facebook Research	
Audio wave	COS 126	
Blood pressure	Wikimedia	CC0 1.0
Object segmentation	PASCAL VOC	
Segmentation for cars	pytorch-semseg	

Credits

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Clustering	Google Developers	
Dimensionality reduction	Wolfram	
PCA Colleges	Ruth Fong	
Anomaly detection	Wolfram	
StyleGAN	arXiv	
Reinforcement learning	Wolfram	
CartPole	Gymnasium	
Cart Pole Games	Andreas De Neve	
Atari Breakout	OpenAI Gym	
AlphaGo	New Yorker	
Rubik’s Cube robot arm	OpenAI	
TossingBot	Google Research	

Credits

Media	Source	License
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US election map	Wikimedia	CC0 1.0
Doctor lung scan	Pexels	
Golden retriever	Emojis.sh	
Doberman	Emojis.sh	
Neuron	Wikimedia	CC BY-SA 4.0
Linear regression	scikit-learn	
Deep neural network	IBM	
Decision tree	Kaggle	